POLICIES OR PREJUDICES? AN ANALYSIS OF ANTISEMITIC AND ANTI-ISRAEL VIEWS ON SOCIAL MEDIA AND SOCIAL SURVEYS

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Abstract
This paper examines the extent to which personal biases affect political views, in the context of how antisemitism influences opinions about Israel. Two empirical analyses are conducted. The first one analyzes social media chatter about Jews and Israel in the UK, revealing a strong, positive relationship between negative chatter about both of them at the daily-location level. In order to establish causality, social media chatter about a "Jewish" football team in the English Premier League (Tottenham) is used as an instrument for negative expressions about Jewish people to explain negativity toward Israel. The second empirical analysis uses the 2016 wave of the German Social Survey, which reveals a strong and robust relationship between several commonly used measures of antisemitic beliefs and holding anti-Israel views. A causal interpretation of this finding is supported by an IV analysis motivated by Voigtländer and Voth who show that Nazi indoctrination during the WWII period had a lifelong impact on antisemitic views. In both analyses, the IV estimates are considerably larger than OLS coefficients. (JEL: P00)

1. Introduction

This paper addresses the general issue of whether an individual's political views are influenced by personal prejudices and biases. This issue is particularly relevant for

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polices that have a disproportionate effect on a minority group. For example, there are many race-neutral reasons for a person to oppose affirmative action, generous welfare programs, or open-border immigration policies. Variation in how people view the role of government and individual rights will naturally lead to variability in support for these kinds of issues. However, it is likely that someone who holds negative opinions about Blacks and immigrants would also oppose affirmative action and unrestricted immigration, respectively. The issue of how to evaluate and weigh the views of individuals suspected of being biased is debated in different strands of academic fields and complicates public policy debates on sensitive issues.

The goal of this paper is to analyze how a person's views about Jewish people affect their views about Israel and the Israeli–Palestinian conflict. The literature on how personal biases affect political views is not extensive, most likely due to the difficulty in establishing causality. Alesina and Stantcheva (2020) present a framework for understanding how biases about immigrants or minorities can shape policy preferences for income redistribution. Huddy and Feldman (2009) argue that the literature on how racial prejudice affects political preferences is unsettled, in large part due to the difficulty in measuring racial animus in surveys. This task is complicated by the potential for "social desirability bias," whereby survey respondents tend to hide their true feelings in order to give socially acceptable responses. As a result, most of the empirical literature focuses on establishing correlations between holding negative views about a group and preferences regarding policies perceived to disproportionately benefit that group.

For example, Sides and Gross (2013) use election surveys to show that negative stereotypes about Muslims are correlated with higher support for the war against Iraq and other aspects of the War on Terror. Fox (2004) uses election surveys to show that negative stereotypes about Latinos and Blacks are associated with preferences against generous welfare policies. Similarly, Gilens (1995, 1996) finds a significant correlation between holding negative opinions about Blacks, and lower support for welfare and means-tested programs. Gilens (1996) argues that negative stereotypes have a profound influence on political preferences. Similar findings appear in Rabinowitz et al. (2009). Stephens-Davidowitz (2014) presents evidence that racial animus, exhibited in Google search queries, reduced the vote-share of Barack Obama in the presidential elections of 2008 and 2012.

Most of the literature on how personal biases affect political views focusses on the issue of immigration. Hopkins (2010) argues that negative attitudes toward immigrants affect policies about border controls. Cattaneo and Grieco (2021) conduct online experiments that manipulate information about immigrants, and show that participants exposed to information about the positive role of immigrants in society display more favorable views about immigrants. Alesina et al. (2023) find that natives hold severe misperceptions about immigrants, and that information designed to correct these misperceptions have little effect on views about redistribution. Relative to our study, these papers are more related to the issues of in-group versus out-group biases, and how ethnic fractionalization affects economic policies and performance (Alesina and La Ferrara 2005).

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Overall, the literature finds suggestive evidence for a link between personal prejudices and political views, but evidence for a causal relationship is scarce. Stronger evidence exists for the effect of party affiliation on political preferences (Gould and Klor 2019).[1] Our paper contributes to this literature by examining the causal effect of antisemitic beliefs on views about Israel. The historical legacy of antisemitism and the salience of the Israeli-Palestinian conflict, both in the political arena and in scholarly inquiry, make this issue both topical and important.

In recent years, reported antisemitic incidents have risen in many Western countries. At the same time, critics of Israel have become more vocal and visible, most notably the Boycott, Divestment, and Sanctions movement. Critics of Israel are keen to make the distinction between antisemitic views and legitimate opposition to the Israeli government and its policies. They argue that false accusations of antisemitism are used to silence criticism of Israel and its policies. On the other hand, Jewish advocacy groups claim that anti-Israeli activists frequently use the language and tropes of traditional forms of antisemitism.[2] They argue that it is antisemitic to focus world attention so intensely on Israel, to hold Israel to a higher standard than other countries, exaggerate criticisms related to Israel, reject Israel's existence and legitimacy, dismiss its security concerns, and blame Israel for problems unrelated to Israeli policies.[3]

1. Gould and Klor (2019) examine whether party switching leads individuals to change their political views on partisan issues, using the increasingly partisan nature of the abortion debate in the 1980s as an instrument for switching parties.

2. Pro-Israel advocates argue that many of the criticisms of Israel echo classical antisemitic tropes. For example, Walt and Mearsheimer (2006) refer to the Israel Lobby as: (1) being a loose organization with a "stranglehold on the US congress"; (2) having unique power over the media, economy, and other institutions; (3) an agent of a foreign power; (4) bearing responsibility for terrorism in the United States and Europe; and (5) leading the US into disastrous wars and policies in the Middle East against its own interests. Pro-Israel advocates point out the similarity of these ideas with a classic antisemitic historical narrative that, as the IHRA (defined below) notes, frequently charges Jews with conspiring to harm humanity, and blames Jews for "why things go wrong."

3. Pro-Israel groups argue that the UN disproportionately condemns Israel. In 2020, according to a report by UN Watch (2020), the UNGA brought 17 resolutions against Israel and 6 against countries in the rest of the world. According to Wikipedia (2021), as of 2013, Israel had been condemned in 45 resolutions by the United Nations Human Rights Council, which represents 46% of all country-specific resolutions passed by the Council, not counting those under Agenda Item 10 (countries requiring technical assistance). Pro-Israeli groups also point to other instances where Israel is blamed for "bad things" unrelated to Jews and Israel. After the 9/11 attacks in 2001, rumors spread that the Israeli Mossad was responsible, and 17 years later (Osborne 2018), a British academic from Sussex University claimed that Israelis blew up the twin towers with help from Zionists inside the US government. The death of George Floyd, a civilian who died while being arrested by Minneapolis policemen, was also linked to Israel. The musician, Roger Waters, claimed that Floyd died due to a technique "taught to the militarized police forces of the United States of America by Israeli experts, who (the Americans) have been flying over to the United States, to teach them how to murder the blacks because they've seen how efficient the Israelis have been at murdering Palestinians in the occupied territories using those techniques . . . and the Israelis are proud of it." Waters proceeded to say that Zionism is an "ugly stain," and that Sheldon Adelson, a prominent donor to Israeli causes, is the "puppet master pulling the strings" of the US President and believes that "only Jewish people are completely human." (See Rutz 2020 about the video of Roger Waters 2020: https://youtu.be/4_yDfyWSDCg.) Statements like these, according to pro-Israel groups, echo antisemitic charges that a mysterious cabal of Jews are controlling the world at the expense of everyone else, and

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In "The Working Definition of Antisemitism" (also called the International Holocaust Remembrance Alliance definition of antisemitism—IHRA definition), which was adopted by 31 countries including 24 in the EU, it states that "criticism of Israel similar to that leveled against any other country cannot be regarded as antisemitic." However, the IHRA statement mentions several examples of antisemitic behaviors related to Israel: accusing Jews of being more loyal to Israel than to their own countries, denying the legitimacy of Israel's existence, applying double standards to Israel by requiring behavior that is not expected or demanded from any other democratic nation, drawing comparisons between Israel and the Nazis, and "using the symbols and images associated with classic antisemitism (e.g. claims of Jews killing Jesus or blood libel) to characterize Israel or Israelis." Consistent with this definition, the President of France, Emmanuel Macron, declared that "anti-Zionism is one of the modern forms of antisemitism" (Lough 2019).

However, according to critics of Israel, charges of antisemitism are used to silence opposing voices and to stifle an honest debate about Israeli policy and US support for it. They claim that this is an effective tactic, since no one wants to be associated with something loathsome like antisemitism.

The goal of this paper is to examine the extent to which antisemitism drives anti-Israel views. The analysis will not reveal the motivations of any particular person. No empirical analysis is capable of doing that. Even if there is no connection between antisemitism and views about Israel, on a personal or aggregate level, a random allocation of views across people on both topics will result in certain people expressing views that are both anti-Israel and antisemitic. The goal, rather, is to empirically examine whether there is a systematic link from antisemitic beliefs to holding very critical views about Israel.

To do this, two empirical strategies are employed—one using social media chatter on Twitter (now known as X) surrounding a "Jewish" football team in the English Premier League, and the second one using social surveys in Germany and Europe.

The first analysis is based on the idea that the Tottenham football team in the English Premier League is widely associated with being a "Jewish" team. The origins of this association date back to its founding in 1882, and are primarily due to its geographic location in the north of London where many Jews live.[4] As described in Wilczyńska (2022), the fans of Tottenham are known as "Yids" (or the "Yid Army") and have embraced many symbols of Jewish identity, even though only about 5% of them are Jewish.[5] Since the 1960's, Tottenham supporters have been frequently met with antisemitic abuse from opposing fans during games, sometimes with references

resemble historical narratives about "blood libels", which link Jews to the murder of non-Jews for their own benefit or enjoyment.

4. https://www.tottenhamhotspur.com/the-club/history/year-by-year/.

5. https://www.thejc.com/lifestyle/features/how-tottenham-became-the-jewish-football-team-1.53784 . Tottenham Hotspur F. C. (2023, April 30). Wikipedia. https://en.wikipedia.org/wiki/Tottenham_Hotspur_F.C.

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about the Holocaust and Nazi gas chambers (Wilczyńska 2022).[6] It is important to note that Tottenham is the only team in England associated with Jews, and is not identified with Israel at all. When queried on ChatGPT about which teams in England are associated with Jews or Israel, Tottenham is the only team mentioned as being associated with Jews, and no team in England is mentioned as being related to Israel.[7]

Based on this unique situation, games against Tottenham, or Tottenham being in the news in general, can be considered a stimulus for attention about Jews. This exogenous variation in exposure to Jewish sentiment is used to explain public expressions about Israel. To do this, we extracted data from 2013 to 2022 from Twitter about tweets mentioning the words Tottenham, Jews, and Israel. A sentiment analysis was conducted to characterize the extent to which each tweet is considered negative, positive, or neutral. Using daily variation across locations, the analysis shows that negative tweets about Jews are highly correlated with negative tweets about Israel. To establish causality, we leverage the finding that social media chatter about Tottenham at the district level, which is associated with the local team playing Tottenham during that week, is positively correlated with negative tweets about Jews. Using negative tweets about Tottenham as an instrument for negative chatter about Jews, the IV analysis shows that negative tweets about Jews increase negative expressions about Israel. In a placebo IV analysis, there is no effect of negative tweets about Jews on negative tweets about China.

The second empirical strategy uses the 2016 wave of the German General Social Survey. To our knowledge, this is the only official governmental survey that contains standard measures of antisemitic views, combined with questions concerning the individual's attitudes about Israel. This analysis is restricted to a sample of native German-born individuals who are neither Jewish nor Muslim, so the conclusions should be considered relevant for those with no first-order connection to the Israeli–Palestinian conflict.

A first pass at the data reveals a strong correlation between holding anti-Israel views and several measures that are commonly used as proxies for antisemitic beliefs, even after controlling for typical demographic variables like age, gender, education, religion, and location. The relationship gets even stronger when the analysis controls for other personal views which may be correlated with the respondent's attitudes toward Jews and Israel, such as: political orientation and opinions about "being a proud German," foreigners, and Muslims. Adding more and more controls strengthens the

6. See these reports from 2007, 2012, and 2023 from The Guardian in Britain https://www.theguardian.com/football/2007/oct/28/newsstory.sport2 https://www.theguardian.com/football/blog/2012/nov/26/west-ham-antisemitic-chants-sickening https://www.theguardian.com/football/2023/jan/18/arsenal-investigating-two-disburbing-anti-semitic-incidents-after-derby.

7. ChatGPT, response to "What football teams are associated with the Jewish community in England," March 20, 2023; and to "What football teams are associated with Israel in England," March 21, 2023. https://chat.openai.com/.

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estimated effect of antisemitic views on anti-Israel sentiments, which supports a causal interpretation of the findings.

To further examine whether a causal relationship exists, the analysis uses an instrument for antisemitic beliefs, which is motivated by Voigtländer and Voth (2015), who show that Germans born during the WWII period were indoctrinated during their childhood by the Nazi party to be anti-Jewish, and these beliefs stayed with them for the rest of their lives. Based on these findings, the IV analysis compares this cohort (born before 1945) with the next cohort (born between 1946 and 1960) as an instrument for holding antisemitic beliefs. The estimated effects are considerably larger than ordinary least squares (OLS) estimates, and following a similar pattern, get stronger when additional controls for demographics and other personal views are added to the specification. A separate analysis with data from the Anti-Defamation League (ADL) on 10 European countries yields similar findings.

There are many papers in recent years related to the causes and consequences of antisemitism.[8] These studies demonstrate that antisemitism, with its significant impacts felt today, has a long and deep-seated history in Europe that precedes the birth of Israel in 1948, the rise of Nazism, or the Zionist movement in the late 19th century.

The empirical literature on the relationship between antisemitism and views about Israel is very thin. Kaplan and Small (2006) examine a survey conducted by the ADL in 10 Western European countries, and found a significant positive relationship between anti-Israeli sentiments and various measures of antisemitic beliefs.[9]

Kempf (2015) examines the "Antisemitism and Criticism of Israel (ASCI)" survey in Germany during 2010, and highlights two main sources of anti-Israeli views: The first is antisemitic views that are frequently correlated with negative views about foreigners, immigrants, and other minority groups; and the second is views that are typically left-wing in nature about human rights, victimization, and pacifism. Kempf (2015) concludes that 26% of the German population criticizes Israel due to antisemitic prejudices, while almost 11% are those who seem to sympathize with Israel and do not take a stance on the Israeli–Palestinian conflict due to latent antisemitism.

8. Voigtländer and Voth (2012) link antisemitism during Nazi Germany to the medieval period (1348–1350) when Jews were often blamed for the Black Death (see also Finley and Koyama 2018). Becker and Pascali (2019) show how the Protestant Reformation in 1517 shifted the map of Jewish persecution in Germany as Jews living in Protestant regions were increasingly exposed to competition in moneylending with the Christian majority, leading to increased antisemitism. Anderson, Johnson, and Koyama (2017) have linked adverse weather shocks to the incidence of Jewish persecutions in European cities from 1100 to 1800. Grosfeld, Sakalli, and Zhuravskaya (2020) also link economic shocks to Jewish pogroms in the Pale of Settlement during the 1800s, especially in areas with political turmoil and where Jews were employed in middlemen occupations. D'Acunto, Prokopczuk, and Weber (2019) show that present-day economic development is lower in German counties where historical antisemitism is higher. Waldinger (2010) and Huber, Lindenthal, and Waldinger (2021) found adverse impacts from the dismissal of Jewish academics and business leaders during the Nazi era on German firms and PhD students.

9. Kaplan and Small (2006) find that individuals with an extreme anti-Israel sentiment are about six times more likely to hold antisemitic views compared with those who do not hold anti-Israel views. This finding persists even after controlling for country of residence, age, religion, income, gender, extent of contact with Jews, attitudes toward people of other races/religions, and views about illegal immigrants. Kaplan and Small (2006) did not focus on the issue of causality (and the direction of causality).

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Kempf (2015) finds that the majority who took a pro-Palestinian stance were not antisemitic, and that critics of Israeli policy could be justified in thinking that charges of antisemitism against them are used to silence their expression. These findings highlight the complexity of the overall issue, and the necessity of untangling how views about Israel and Jews are related to left- and right-wing political leanings, as well as views about other minority groups.

Recent experiments conducted in person and online found a significant relationship between antisemitic and anti-Israel sentiments, and highlight how this correlation varies with the level of knowledge about Israel and the extent to which participants believe lying can be detected.[10] These findings highlight concerns about respondents under-reporting their antisemitic leanings in survey data. Overall, our study is the first to use social media and survey data to analyze the causal relationship between antisemitism and views about Israel, by using a new and distinct IV strategy for each source of data.

2. Twitter Data and OLS Results

Our first empirical strategy analyzes social media chatter about Jews and Israel, with expressions about a "Jewish" football club in the English Premier League (Tottenham) used as an instrument for chatter about Jewish people. We collected Twitter data for tweets, geocoded by location in the UK, mentioning the words: Israel, Jew, and Tottenham (including variations). The sample period starts at the beginning of 2013 and extends to the end of 2022. The number of tweets per day mentioning each keyword was then aggregated to the location level (355 districts). Dates with military hostilities between Israel and Palestinian factions were dropped from the sample (see the Online Appendix for details). The main sample consists of 1,253,505 observations (3,531 days for each of the 355 districts).

A sentiment analysis was conducted to characterize whether each tweet was positive, neutral, or negative. This analysis was performed with "TweetNLP", which uses the "Robustly Optimized BERT-Pretraining Approach" (RoBERTa) architecture (Camacho-Collados et al. 2022). This process produces the percentage chance that

10. Cohen et al. (2009) conducted experiments on 151 students in a Social Psychology course at Rutgers University, and found that an increasing salience of mortality (which is intended to prompt more honest revelations of world views) led students to express more antisemitic views and negative sentiments towards Israel (but not China). They also found that reported levels of antisemitism were higher when the subjects were led to believe that they would be caught lying about their overall views—relative to the case where subjects were told that the survey concerned their "prejudices." Cohen (2009) used a "bogus pipeline treatment" whereby participants were told that the survey was employing techniques that psychologists use to detect when people are lying in order to give a positive impression of themselves. Beattie (2016) conducted an internet survey on M-Turk, and found support for the finding in Kaplan and Small (2006) that antisemitism is highly correlated with extreme criticism of Israel and inversely related to knowledge about Israel. Beattie (2016) also found that higher mortality salience did not evoke greater antisemitism (and Islamophobia). The M-Turk sample in Beattie (2016) is not representative, and is disproportionately white, young, and liberal.

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TABLE 1. Summary statistics of Twitter data in the UK, 2013–2022.

	Means		
	All days and locations	Tottenham game window (4 days before and after)	
		All locations	Locations of the opposing team
	(1)	(2)	(3)
Total tweets per day by district about:			
Israel	0.112	0.113	0.282
Israel—with positive sentiment	0.015	0.015	0.038
Israel—with negative sentiment	0.053	0.054	0.130
Jews	0.087	0.088	0.233
Jews—with positive sentiment	0.015	0.016	0.045
Jews—with negative sentiment	0.038	0.038	0.097
Tottenham	0.163	0.192	0.745
Tottenham—with positive sentiment	0.048	0.056	0.222
Tottenham—with negative sentiment	0.035	0.042	0.137
Observations	1,253,505	867,620	48,750

Notes: The sample is taken from Twitter in the years 2013–2022, excluding days of hostilities. The data include tweets with information on location (355 districts) in the UK. The main sample consists of 1,253,505 observations (3,531 days from 2013 to 2022 for each of the 355 districts). Locations of the opposing team are defined to be districts that are within 25 km of the opposing team's stadium that is playing Tottenham within that 8-day window.

a tweet is of a positive, neutral, or negative nature (all three add up to 100%). (See examples in the Online Appendix). To aggregate the Twitter data by day and location, the total number of tweets (about Tottenham, Jews, or Israel) is added up. To obtain the total amount of negative sentiment for Jews (or Israel or Tottenham), the percent of each tweet about Jews that is negative is added up by day and location. For example, if there are three tweets about Jews, with two that are 40% negative and one that is 60% negative, then the three tweets would count as 1.4 negative tweets. The total positive sentiment by day and location is created similarly. Further details about the data extraction and variable construction are presented in the Online Appendix.

Summary statistics for tweets by day and location are presented in Table 1. There are more tweets per day using the word "Tottenham" relative to tweets mentioning "Israel" or "Jew." According to the sentiment analysis, most tweets about all three topics are neutral in character. The tweets about Tottenham that are not neutral tend to be on the positive side, while tweets about Jews are two times more likely to be negative rather than positive, and negative sentiment about Israel is almost four times as common as positive expressions. The second column presents the means for the main variables during a window around days that Tottenham plays a game (four days before, the day of the game, and the four days after). Column (3) displays the means during this window but only in areas close to where the opposing team plays their home

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games. Table 1 shows that social media chatter is much higher for all three topics in areas near the opposing team around game day. This pattern shows heightened interest in the game, and also in matters related to Jews and Israel.[11]

The empirical strategy for the Twitter analysis leverages variation by day and location in negative social media attention about Jews in order to explain negative sentiment toward Israel. The main estimating equation is

$$Israel\ Negative_{jt} = \beta_0 + \beta_1 Jews\ Negative_{jt} + u_j + \delta_t + \varepsilon_{jt}, \quad (1)$$

where $Israel\ Negative_{jt}$ is the total number of negative tweets about "Israel" (defined above) in district j on day t; $Jews\ Negative_{jt}$ is the total number of negative tweets about Jews (defined above) in district j on day t; μ_j is a fixed-effect unique to district j (355 districts); and δ_t represents a series of time fixed-effects for each year, month of the year, and day of the week. The error term is represented by ε_{jt}.

The main parameter of interest is β_1, which represents how much negative sentiment about Jewish people affects negative expressions about Israel. It is important to note that the analysis is not testing whether an individual who posts about Jews is also posting about Israel. In fact, only 2.5% of the people who post about Israel also post something on the same day mentioning Jews. Rather, social media activity at the local level can be regarded as a proxy for the overall local sentiment toward Jews and Israel. If more people are expressing certain thoughts about Jews or Israel on social media on a given day in a certain locality, this activity is likely to be driven by events or news items that are known more generally in the local population, even by those that are not taking the time to post on social media. In this manner, the empirical strategy examines whether geographic variation in the salience of Jews increases negative sentiment toward Israel. In order to establish a causal nature of this relationship, social media chatter about Tottenham will be used as an instrument for the local salience of Jews.

Table 2 presents the OLS relationships between social media postings about Israel, Jews, and Tottenham. Column (1) shows that expressions about Israel are very positively correlated with tweets about Jewish people. The next few columns break down the tweets by positive and negative sentiment. Positive tweets about Israel are related to positive and negative tweets about Jews, but the relationship is much stronger for positive tweets about Jews. The reverse is true for negative sentiment about Israel, which is related only to negative sentiment about Jews, with no significant coefficient on positive Jewish sentiment. The most pronounced relationship revealed in these columns is that negative chatter about Israel in the UK on social media is very strongly related to negative chatter about Jews, and this is not sensitive to controlling for positive chatter about Jews or not.

In columns (5)–(7) in Table 2, positive and negative chatter about Jews is found to be significantly related to negative tweets about Tottenham. Positive tweets about

11. The higher means could also be due to the selection of districts which contain teams that play against Tottenham. Localities with professional teams may have a higher population density, and this may lead to higher overall social media use. The differences across localities in social media use will be neutralized in the analysis with location fixed-effects.

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TABLE 2. OLS relationships between tweets about Israel, Jews, and Tottenham.

	Israel (1)	Israel positive (2)	Israel negative (3)	Israel negative (4)	Jews positive (5)	Jews negative (6)	Jews negative (7)	Israel positive (8)	Israel Negative (9)	Israel Negative (10)	China Negative (11)	China Negative (12)
Jews	0.305***											
	(0.048)											
Jews—positive		0.045***	0.046									
		(0.012)	(0.035)									
Jews—negative		0.019***	0.294***	0.297***								
		(0.005)	(0.046)	(0.047)								
Tottenham—positive					0.003**	0.007		0.001	0.008		−0.002	
					(0.001)	(0.005)		(0.001)	(0.006)		(0.002)	
Tottenham—negative					0.008***	0.016***	0.020***	−0.000	0.010*	0.015*	0.002	0.001
					(0.002)	(0.003)	(0.004)	(0.002)	(0.006)	(0.008)	(0.004)	(0.005)
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,253,505											

Note: Robust standard errors, clustered by district are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each column represents a separate regression. Each observation in the sample represents the total number of tweets by day (3,531 days) and location (355 districts). The sample is described in the notes to Table 1. Every specification controls for fixed-effects for 355 locations, year, month of the year, and day of the week.

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Tottenham are related to positive Jewish expressions, but the effect is much smaller in magnitude compared with negative tweets about Tottenham. Regarding negative tweets about Jews, only negative tweets about Tottenham is significant, with smaller and insignificant coefficients for positive tweets about Tottenham. The main pattern here is that negative tweets about Tottenham and Jews are strongly related, and this is essentially the first stage of the IV analysis of equation (1) that examines the causal nature of the estimated effect of negative chatter about Jews on negative expressions of Israel in column (4).

In columns (8)–(10) in Table 2, positive tweets about Israel are found to be unrelated to positive or negative tweets about Tottenham, but negative tweets about Israel are associated with negative tweets about Tottenham. This finding is consistent with the idea that negative tweets about Tottenham increase negative expressions of Israel indirectly by increasing the salience of Jewish issues.

Overall, Table 2 shows that when Tottenham arouses attention in social media, negative chatter about Jews and Israel arise in its wake. The last two columns of Table 2 present a placebo-like analysis using social media chatter about China.[12] (Cohen et al. 2009 also used views about China as a placebo to study how antisemitism affects stances toward Israel.) The table shows that negative expressions about China are not correlated with any kind of expressions about Tottenham. This pattern supports the idea that social media chatter about Tottenham increases the salience of matters related to Jews, but does not provoke expressions about other countries or ethnic groups. This finding supports the IV assumptions that tweets about Tottenham affect expressions about Jews, but have no direct effect on social media attention about any specific country—particularly one associated with a minority group.

Online Appendix Table A.1 shows that social media about Tottenham is not random, but rather is picking up news related to the team. The first two columns show that negative chatter about the team is significantly higher in days near one of their games, and even more so in areas close to the opposing team. The table also shows that negative tweets about Jews and Israel increase near game days for Tottenham. Defining what it means for a district to be close to the opposing team is complicated by the high concentration of teams in London (see the Online Appendix for details). For that reason, we regard the district-level chatter about Tottenham for each day as the best measure of Tottenham's salience at that particular moment in time and place. But, even with this caveat, Online Appendix Table A.1 shows that chatter about Tottenham is related to the timing and location of their games. In contrast, negative tweets about China decline around game days for Tottenham, most likely due to a crowding-out effect as social media focusses more on sports in general during these days. This finding highlights how the increase in social media attention surrounding Jews and Israel close to game days for Tottenham occurs despite heightened interest in sports relative to politics during these days.

12. We note that Tottenham did not have players from China or Israel during the sample period.

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3. Twitter Data and IV Results

Table 2 displayed a strong OLS effect of negative tweets about Jews on negative tweets about Israel. This section examines the causal nature of this relationship by using social media chatter about Tottenham as an instrument for negative social media attention about Jews. Table 3 presents the main results from this analysis using different ways of specifying Tottenham chatter as the instrument.

The first column of Table 3 uses negative tweets about Tottenham as an instrument for negative tweets about Jews. The estimated effect of a negative tweet about Jews on negative tweets about Israel is 0.748, which is statistically significant and more than double the size of the corresponding OLS coefficient of 0.297 in Table 2. The F-statistic for the excluded instrument in the first stage is above 25, which is sufficiently strong by conventional standards. This finding shows not only that negative social media attention about Jews is correlated with negative chatter about Israel in the UK at the local and daily level, but also that there is a strong and positive causal effect of negative sentiment about Jews on negative tweets about Israel. Moreover, the strong correlation in the OLS analysis between social media chatter about Jews and Israel understates the size of the causal effect. This pattern is a sign that people may be revealing their antipathy for Jews through negative expressions about Israel.

Table 3 also examines alternative ways of specifying the instrument for negative tweets about Jews. These alternatives include adding positive tweets about Tottenham as an instrument, and using the window around game days for Tottenham in general and specifically in the location of the opposing team. A clear pattern emerges that the estimated effect of negative tweets about Jews has a large and statistically significant effect on negative tweets about Israel, and this is robust to the different ways of specifying the instrument. All of the estimated effects are much larger than the OLS estimate, except for a similarly sized coefficient using only the dummy variables for each day in the window surrounding the game day for Tottenham. (This is also the only specification with an F-statistic in the first stage which is not sufficiently strong by conventional standards.)

Table 3 also examines the effect on positive tweets about Israel. Column (7) reveals an insignificant estimate of negative tweets about Jews on positive tweets about Israel. This result stands in sharp contrast to the results for negative tweets about Israel—demonstrating that negative tweets about Jews have a very strong positive effect on negative tweets about Israel, with little influence on positive tweets about Israel.

Table 3 presents a placebo analysis with negative tweets about China as the outcome. Despite the strong first stage F-statistics, the estimated causal effect of negative social media attention about Jews on China is essentially zero. Negative chatter about Jews has a strong effect on negative chatter about Israel, but no effect on China.[13]

13. Online Appendix Table A.2 attempts to separate the effects of negative and positive chatter about Jews on tweets about Israel, however the instruments are not sufficiently strong to adequately separate these effects. As discussed below, social media about Tottenham is likely to spur both negative and positive

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TABLE 3. IV results for tweets about Israel.

	Israel negative						Israel positive	China Negative	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Jews negative	0.748**	0.839**	1.493***	0.317**	0.879***	0.765***	0.042	0.047	−0.025
	(0.308)	(0.343)	(0.462)	(0.142)	(0.331)	(0.290)	(0.083)	(0.233)	(0.253)
Instruments for Jews negative									
Tottenham negative	Yes	Yes			Yes	Yes	Yes	Yes	Yes
Tottenham positive		Yes			Yes	Yes			Yes
Tottenham game window			Yes		Yes				
Opposing location * Tottenham game window			Yes		Yes				
Tottenham game window (daily effects)				Yes		Yes			
Opposing location * Tottenham game window (daily effects)				Yes		Yes			
F-statistic for excluded instruments in first stage	25.83	16.75	14.20	4.18	14.13	6.81	25.83	25.83	16.75
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,253,505	1,253,505	1,246,760	1,246,760	1,246,760	1,246,760	1,253,505	1,253,505	1,253,505

Note: Robust standard errors, clustered by district, are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each column represents the second stage from a separate 2SLS regression. Each observation in the sample represents the total number of tweets by day (3,531 days) and location (355 districts). The sample is described in the notes to Table 1. Every specification controls for fixed-effects for 355 locations, year, month of the year, and day of the week. "Tottenham Game Window" is a dummy variable equal to one for the four days before Tottenham plays, the day of Tottenham's game, and the four days after Tottenham's game. "Opposing Location" is a dummy variable equal to one if the the district is within 25 km of the opposing team's stadium that is playing Tottenham (regardless of whether Tottenham is playing at home or away). Variables with "Daily Effects" include dummy variables for each of the days within the indicated window.

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TABLE 4. Main Twitter results using observations at the weekly level.

	OLS				IV		
	Israel (1)	Israel positive (2)	Israel negative (3)	Israel negative (4)	Israel positive (5)	Israel negative (6)	Israel negative (7)
Jews	0.595***						
	(0.112)						
Jews—positive		0.078**	0.161				
		(0.031)	(0.132)				
Jews—negative		0.038***	0.547***	0.561***	0.198***	0.955**	1.114**
		(0.012)	(0.101)	(0.108)	(0.065)	(0.388)	(0.477)
Instruments for Jews-Negative							
Tottenham Negative					Yes	Yes	Yes
Tottenham Positive							Yes
R^2	0.303	0.094	0.258	0.256			
Time FE (year and week)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	180,340						

Note: Robust standard errors, clustered by district, are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each observation in the sample represents the total number of tweets by week and location (355 districts). The sample is described in the notes to Table 1. Every specification controls for fixed-effects for 355 locations, year, and week of the year.

Overall, the OLS analysis reveals a strong positive correlation at the location and daily level between negative tweets about Jews and Israel. The IV analysis reveals a much stronger causal relationship of negative sentiment about Jews on negative tweets about Israel. One empirical concern is that there are many observations at the day and location level with zeros as the outcome variable (i.e., there are no tweets about Israel on many days in many localities in the sample). We address this issue in a few ways. Table 4 presents the main results after aggregating the data to the weekly level by location. There are far fewer zeros at the weekly level compared with the daily level. The results in Table 4 are very similar to those in previous tables. The most prominent finding in the OLS analysis is the strong effect of negative tweets about Jews on negative tweets about Israel. The IV analysis reveals a small effect of negative tweets

expressions about Jews, making it difficult to identify each effect separately. However, if there is a pattern, Online Appendix Table A.2 suggests that the effect of negative tweets about Jews has a stronger effect than positive tweets about Jews. The sensitivity of the results to using a 25 kilometer radius to define what qualifies as the "opposing team" is displayed in Online Appendix Table A.3. The findings are very similar if we use a 15 or 20 kilometer radius.

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about Jews on positive tweets about Israel, but a much larger effect on negative tweets about Israel. The IV estimates of the latter are very similar to the IV results at the daily level of aggregation (Table 3). All of these results, in terms of size and significance, are similar to those obtained after aggregating the data to the monthly level, as seen in Online Appendix Table A.4. There are even fewer zeros at the monthly level of aggregation, compared with the weekly and daily levels. Therefore, there is little evidence that the prevalence of zeros is affecting our conclusions.

The analyses at the weekly and monthly level also provide support for the idea that measurement error is biasing the OLS results at the daily level toward zero. The OLS coefficient on negative tweets about Jews in Table 2 is 0.297, compared with 0.561 at the weekly level (Table 4) and 0.670 at the monthly level (Online Appendix Table A.4). The specification at the daily level is using chatter about Jews to explain tweets about Israel within the same day, but it may take a few days for these expressions and reactions to unfold. Therefore, there is likely to be less measurement error at the weekly or monthly level of aggregation, and this could explain why the estimated effects get much larger and move toward the magnitudes obtained in the IV analysis.

In order to fully model the count-data nature of the outcome variable, Online Appendix Table A.5 presents estimates from a Poisson model with fixed-effects. The coefficients cannot be compared in terms of the magnitudes to previous tables, but the sign and significance are very similar. Negative tweets about Tottenham increase negative tweets about Israel, and negative tweets about Jews increases both negative and positive tweets about Israel, but the effect on negative sentiment about Israel is larger.

Table 5 presents our main OLS and IV results using all tweets about Jews as the treatment variable, and all tweets about Tottenham as the instrument. The OLS and IV results show that increasing social media attention to all matters related to Jews increases negative expressions about Israel, but has very little effect on positive sentiments toward Israel. These findings are robust to using all postings about Tottenham as the instrument for tweets about Jews, or only the positive and neutral tweets about Tottenham. Therefore, it is not the case that negative "trash talk" about Tottenham by opposing fans is driving our results.[14] Table 5 shows that an increased local exposure to anything related to Jews, as instrumented by overall exposure to Tottenham, is causing negative sentiment toward Israel. Table 5 also demonstrates that the main findings are not driven by potential classification errors in the sentiment analysis of tweets about Jews and Tottenham, as similar results are found using all tweets about Jews and Tottenham.

Overall, the OLS and IV findings point to a strong causal link between social media expressions about Jews on chatter about Israel. It should not be surprising that there is an effect on both positive and negative attention about Israel, since negative tweets about Israel from some people are likely to attract positive responses from

14. It is also important to note that opposing fans are more likely to use the word "yid" in derogatory remarks about Tottenham, as opposed to the word "Jew"—which is the keyword that we searched for. Also, although opposing fans may engage in "trash talk" chanting and behavior around gametime with a group of fellow fans, our sample only includes individuals that take the time to personally write a posting on social media.

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TABLE 5. Main Twitter results using all tweets about Jews and Tottenham.

	Israel positive				Israel negative			
	OLS	IV			OLS	IV		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Jews	0.020***	0.051**	0.060**	0.017	0.159***	0.416**	0.444**	0.309**
	(0.004)	(0.026)	(0.028)	(0.034)	(0.027)	(0.188)	(0.210)	(0.125)
Jews—negative	0.022***	0.123*	0.143*	0.042	0.297***	0.996**	1.035**	0.748**
	(0.005)	(0.068)	(0.073)	(0.083)	(0.047)	(0.468)	(0.504)	(0.308)
Instruments for Jews or Jews-negative								
Tottenham—all		Yes				Yes		
Tottenham—positive			Yes				Yes	
Tottenham—neutral			Yes				Yes	
Tottenham—negative				Yes				Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,253,505							

Note: Robust standard errors, clustered by district, are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each coefficient represents a separate regression with the dependant variable indicated in the column heading and the treatment variable indicated by the row. The IV results represent the second stage treatment estimate. Each observation in the sample represents the total number of tweets by day (3,531 days) and location (355 districts). The sample is described in the notes to Table 1. Every specification controls for fixed-effects for 355 locations, year, month of the year, and day of the week.

others in Israel's defense. In other words, increased social media attention about Jews is likely to spur discussions and arguments about Jews and Israel, resulting in spikes in positive and negative chatter about both topics.[15] However, the most prominent pattern by far in the analysis is the very strong effect of negative chatter about Jews on negative tweets about Israel. This finding is consistent with the idea that negative expressions about Israel are very influenced by negative sentiment about Jews.

4. The German Social Survey Data

This analysis uses data from the German General Social Survey (ALLBUS) from 2016.[16] Voigtländer and Voth (2015) use the 1996 and 2006 waves of the ALLBUS

15. Also, by definition, negative and positive sentiments about Israel (or Tottenham or Jews) have a positive correlation. This is due to the way these variables are constructed. Every tweet is characterized by how much it is positive, neutral, and negative. So, each additional tweet by definition increases the amount of negative, neutral, and positive sentiment on that day.

16. GESIS—Leibniz-Institut für Sozialwissenschaften (2018). German General Social Survey—ALLBUS 2016. *GESIS Data Archive, Cologne. ZA5252 Data file Version 1.0.0,* https://doi.org/10.4232/1.12837.

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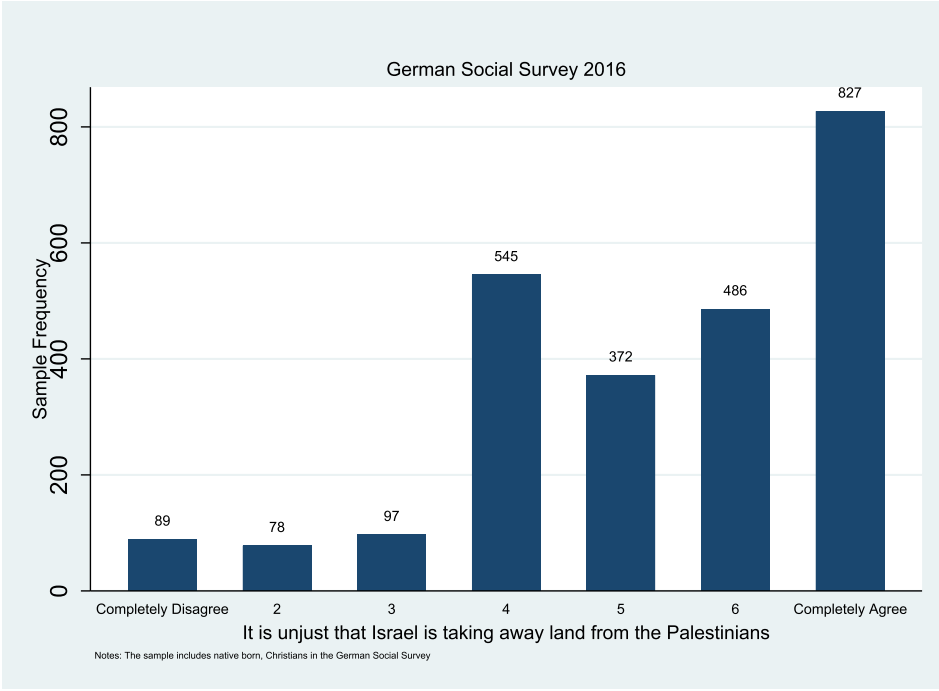


FIGURE 1. Frequency of views about "Israel Taking Palestinian Land".

surveys to demonstrate that Nazi indoctrination during the 1930's and 1940's had a lasting impact on antisemitic beliefs many decades later. The 2016 wave contains two questions about the respondent's views toward Israel. (These questions appear only in the 2016 wave of the ALLBUS survey.) The ALLBUS survey in 2016 represents a unique opportunity to study views about Israel and Jewish people from a survey conducted by an official governmental statistical agency.

The sample is restricted to individuals born in Germany between 1930 and 1990, and responded that their religious affiliation is unaffiliated or Christian. These sample restrictions are designed to exclude groups, such as Jews and Muslims, which may have first-order religious or ethnic considerations that could influence their views toward Israel and its relationship with the Palestinians. The two questions in the 2016 ALLBUS survey regarding Israel ask respondents to indicate how much they agree or disagree with the following statements: (1) "It is unjust that Israel is taking away land from the Palestinians"; and (2) "Given Israel's policies, I can well understand that people are against the Jews." Respondents were asked to rank their level of agreement from 1 ("completely disagree") to 7 ("completely agree").

The distribution of views in the sample described above is displayed in Figures 1 and 2, and in the descriptive statistics in Table 6. Figure 1 shows that most Germans in the sample agree with the first statement, with very few respondents showing any level

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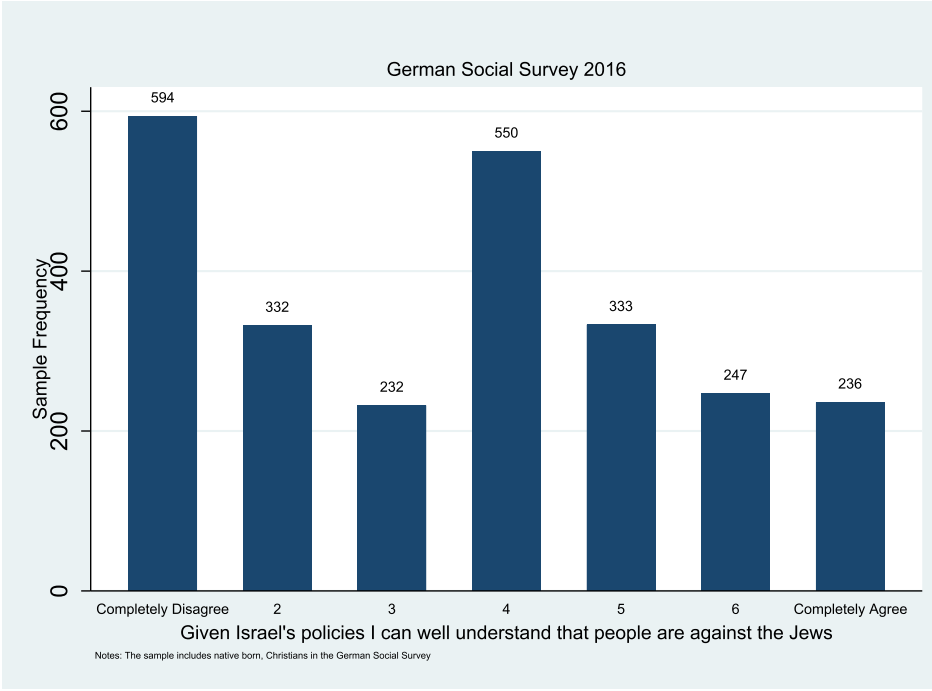


FIGURE 2. Frequency of views about Israeli policy.

of disagreement. The mean ranking of this statement is 5.3, which tilts heavily into the "agreement" portion of the distribution. There is variation in the responses in the upper half of the distribution though. It should be noted that someone may disagree with this statement for two different reasons: They may think it is just for Israel to take Palestinian land, or they may believe that Israel is not taking Palestinian land.

The second statement regarding views about Israel ("Given Israel's policies, I can well understand that people are against the Jews") has a much lower mean (3.55) and more variation than the first statement, showing that a much larger proportion of the sample disagrees that Israel's policies justify anti-Jewish sentiments. Figure 3 shows the relationship between the two different statements about Israel. As the figure shows, there is a direct monotonic relationship between the two, which indicates that both are eliciting similar sentiments about how the person regards Israel. For the purposes of this study, the first statement is a cleaner expression of one's views about Israel in the sense that it is not mixed with issues related to Jews. Nonetheless, neither of these survey questions are ideal, but the consistency across both measures indicates that they are reflecting accurate personal sentiments regarding Israel. It is hard to imagine that these variables are not revealing meaningful variation in how respondents feel about Israel. Therefore, the analysis will concentrate on the first question as the main outcome of interest, but both measures will be used to check the sensitivity of the main findings.

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TABLE 6. Summary statistics of the German Social Survey 2016.

	All		Sample: Israel unjust = 7		Jewish influence =7
			Jewish influence >3	Jewish influence <=3	
	Mean (1)	Standard deviation (2)	Mean (3)	Mean (4)	Mean (5)
Scale from 1–7: Completely disagree = 1 and completely agree = 7					
Israel—unjust to take Palestinian land	5.30	1.62	7.00	7.00	6.05
Israeli policy justifies antisemitism	3.55	1.97	5.25	3.76	5.47
Jews—too much influence in the world	3.09	1.92	5.38	1.51	7.00
Jew are not without blame for persecution because of their behavior	2.26	1.72	3.14	1.73	3.94
Many jews take advantage today of the Holocaust	3.92	2.04	5.47	3.28	5.96
I'm not ashamed of German crimes against Jews	2.20	1.76	2.27	1.85	2.75
It would be unpleasant to have a Jewish neighbor	3.45	1.37	3.92	3.07	4.36
Foreigners enrich German culture	4.18	1.84	3.74	4.70	3.24
Islam fits in with German society	2.75	1.77	2.13	3.05	1.79
First factor of five views about Jews	0.00	0.82	0.81	−0.51	1.35
Mean of five views about Jews	2.99	1.19	4.04	2.28	4.82
Political spectrum (left to right from 1 to 10)	5.06	1.72	5.38	4.63	5.48
Very proud German (dummy)	0.27	0.45	0.33	0.28	0.39
No religion (dummy)	0.43	0.49	0.48	0.45	0.49
College graduate	0.29	0.45	0.18	0.36	0.09
Age	54.59	15.24	60.66	56.82	62.48
Cohort: born 1930–1945	0.18	0.39	0.32	0.23	0.36
Cohort: born 1930–1960	0.47	0.50	0.64	0.51	0.71
Male	0.52	0.50	0.60	0.56	0.70
Observations	2,336	2,336	382	342	165

Notes: The sample is taken from the German General Social Survey (ALLBUS) from 2016, and is restricted to individuals born in Germany between 1930 and 1990 that categorized their religious affiliation as unaffiliated or Christian. The sample is restricted to observations where all the main variables used in the empirical analysis are non-missing (i.e., column (7) of Table 7). The third and fourth columns are restricted to the sample of those that responded with "7" to the question about whether Israel is "unjust to take Palestinian land." Among this sample, column (3) is restricted further to those that responded with more than "3" to the question about whether Jews have too much influence in the world, and column (4) is restricted to those that answered "3" or less to the same question.

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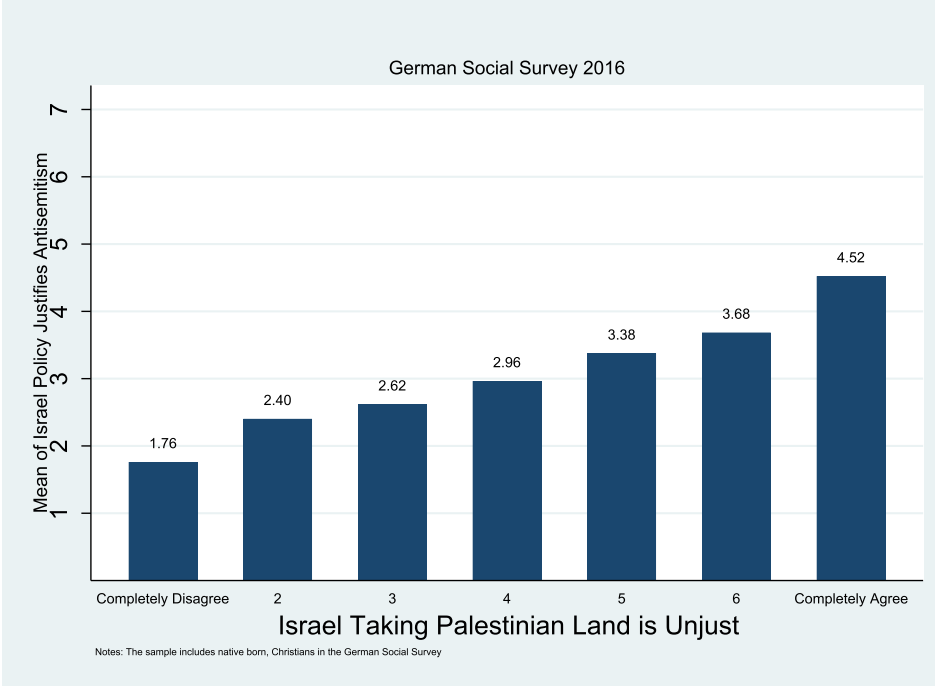


FIGURE 3. Two views on Israel.

The 2016 ALLBUS survey also contains views about Jews that are commonly used to formulate proxies for antisemitic beliefs (see Voigtländer and Voth 2015). Again, respondents were asked to rank how much they disagree or agree with certain statements about Jews from 1 to 7. Table 6 presents descriptive statistics of these views, after re-coding some of them so that a higher number is indicative of a more negative view toward Jewish people. In particular, respondents were asked how much they agree that: (1) Jews have too much influence on the world; (2) Jews are not without blame for their persecution because of their behavior; (3) Many Jewish people try to take personal advantage today of what happened during the Nazi era and make Germans pay for it; (4) I'm not ashamed that Germans have committed so many crimes against the Jewish people; and (5) It would be unpleasant to have a Jewish neighbor.

According to Table 6, the means for all of these views lie toward the "disagree" side of the spectrum (indicating a favorable attitude toward Jews) except for the question related to whether Jews take advantage of the Holocaust today. The distribution of responses for the first view, regarding whether Jews have too much influence on the world, is displayed in Figure 4. The most common response to this question is "total disagreement" that Jews have too much influence, but there is considerable variation. This question is similar to the thesis propagated by the infamous antisemitic tract,

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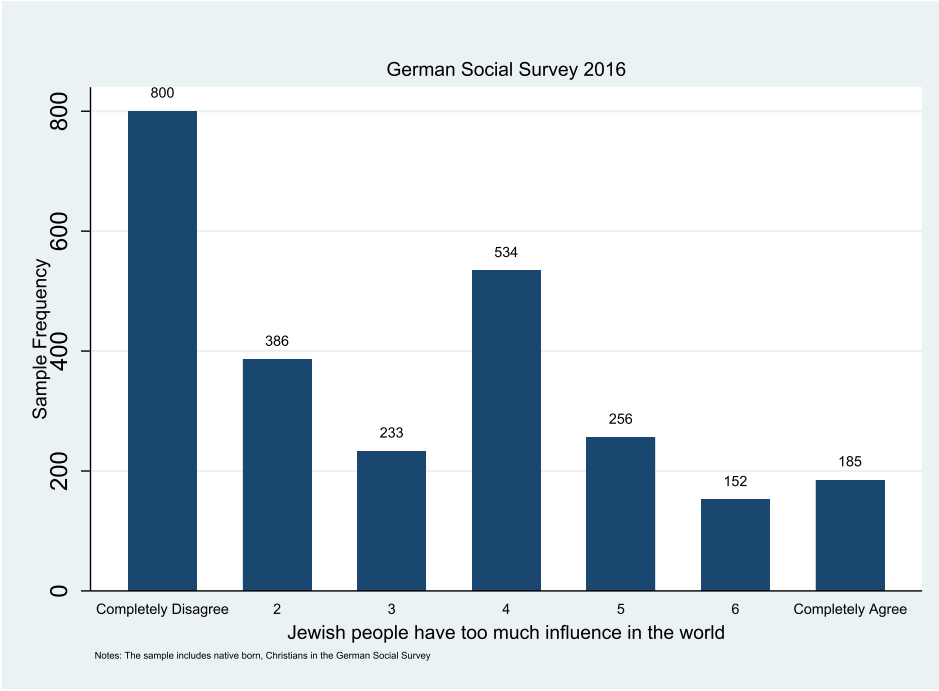


FIGURE 4. Frequency of views about Jews having influence.

the Protocols of the Elders of Zion, which fabricated a secret Jewish cabal for world domination. For this reason, the analysis focusses on this view as the main indicator for antisemitic beliefs, but the other views will also be examined for robustness.[17] In addition, the analysis will use two different composite measures of all five views: the mean of all five views and the first factor after performing factor analysis on all five views (which produced factor loadings consistent with the first factor increasing with anti-Jewish sentiment).[18]

Figures 5 and 6 present the relationship between views toward Jews and about Israel. Both figures display a strong correlation between thinking that Jews have too much influence and both measures of anti-Israeli sentiment. This correlation

17. Voigtländer and Voth (2015) show that this measure of antisemitism, in surveys from 1996 and 2006, is significantly related to Nazi indoctrination during the respondent's youth, and is related to the district-level of support for antisemitic parties before 1914. These results demonstrate that this measure is indeed picking up antisemitic sentiments. The first finding is used as the basis of the IV analysis presented later. The second finding, regarding voting patterns, is statistically significant in Voigtländer and Voth (2015). But, with t-statistics around 2.5, these results are not strong enough to be used as a first-stage in an IV analysis.

18. The factor loadings on the first factor (in order of how the variables are listed in Table 6) are: 0.67, 0.62, 0.65, 0.28, and 0.43. Therefore, the one variable which is heavily discounted (with a factor loading of 0.28) is the variable for "Not Ashamed of German Crimes against Jews."

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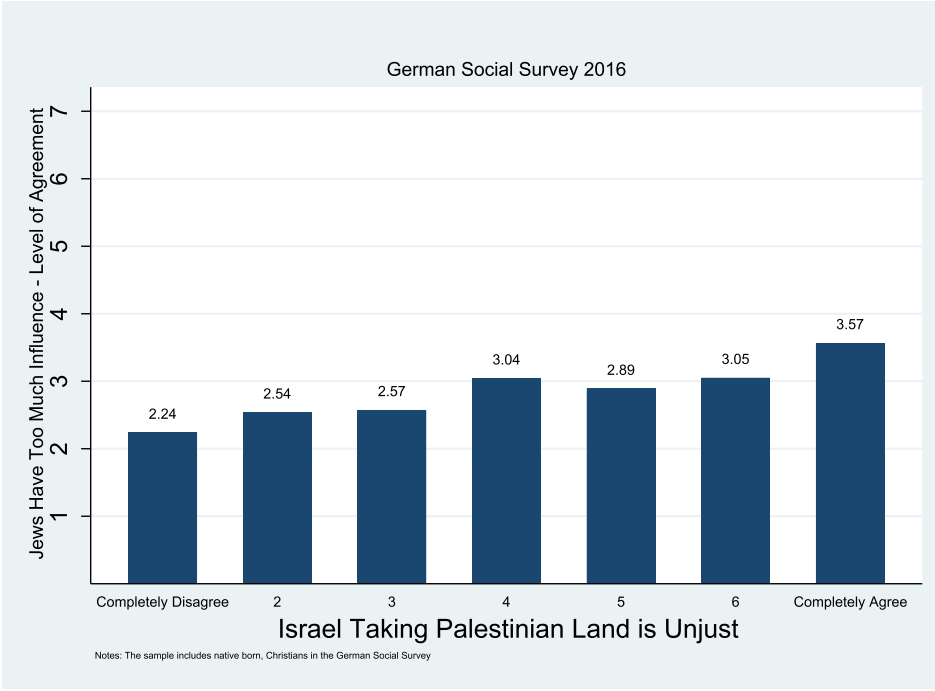


FIGURE 5. Views on Jewish influence by views on Israel.

is noteworthy for many reasons, and debunks the idea that views about Jews are generally unconnected with views toward Israel. However, the goal of the empirical analysis is to examine whether antisemitic beliefs cause a person to hold anti-Israel views. To do this, the analysis will examine whether this relationship is spurious in the sense that both are correlated with a person's overall political and ideological worldview. Table 6 presents summary statistics for other variables that will be used to control for a person's general ideology: age, gender, education, and views about foreigners, Islam, political leanings, religiosity, and German patriotism. These latter characteristics will be used to examine whether the correlation between views on Jews and Israel is causal, and this sensitivity analysis will be augmented by an IV strategy.

Before presenting this analysis, it is worth examining the people with the most negative views about Israel—the 32% of the sample with the most anti-Israel response to the question about whether it is unjust for Israel to take Palestinian land ("completely agree" in Figure 1). Columns (3) and (4) of Table 6 display the characteristics of this group, while breaking it down into those that expressed some disagreement with the idea that Jews have too much influence in the world (≤ 3 in Figure 4) versus those that do not express disagreement with this idea (more than three in Figure 4). Most of this group (65%) do not express disagreement with the idea that Jews have too much influence (column (3)), and their responses to the other questions about Jews show a

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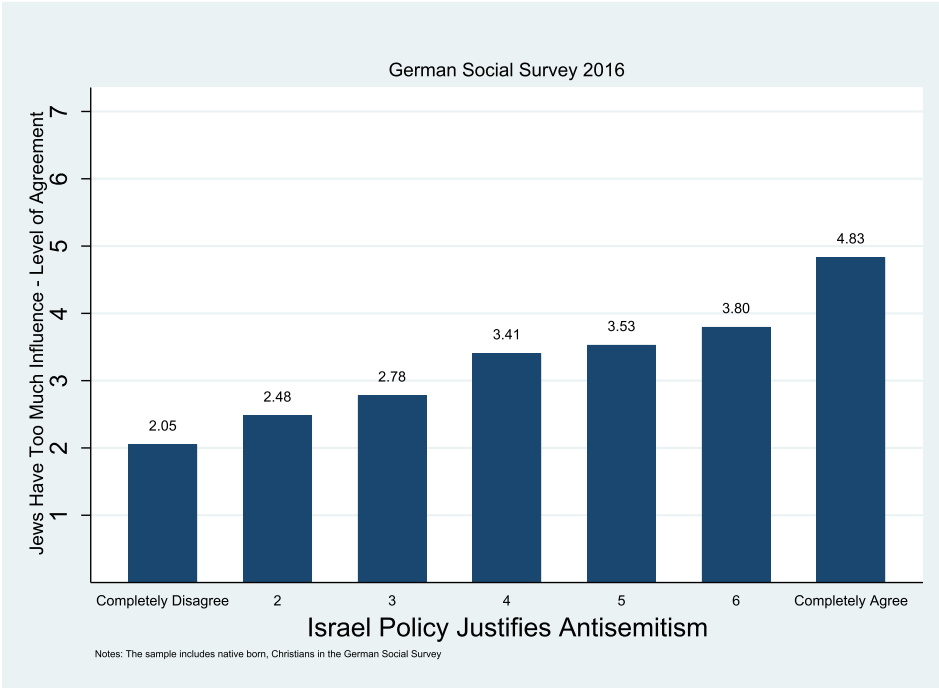


FIGURE 6. Views on Jewish influence by views on Israeli policy.

lot more antipathy than those in column (4) who have very negative views about Israel, but express some disagreement that Jews have too much influence. Respondents with negative views about Israel and Jews (column (3)) are much more negative across all measures of feelings about Jews relative to the whole sample (column (1)) or those that are very negative about Israel but express some disagreement about Jews having too much influence.

The group in column (4) of Table 6 is interesting because they are very negative about Israel but do not show significant signs of antisemitism. This group is roughly half of the third of the entire sample with the most hostile views of Israel, and they are different from those that are negative about both Israel and Jews in several dimensions. They are much more likely to have a college degree (36% vs. 18%), to be on the left side of the political spectrum, to have more favorable views about foreigners and Islam, and are about 4 years younger on average. These patterns suggest that anti-Israel sentiment is stronger for older people who are more right-wing and less open to Jews and other ethnic groups, and with younger people who are more left-wing, educated, and more accepting of minorities. It could also be the case that younger people are more cautious and politically-correct when queried about sensitive opinions which could be considered antisemitic or racist. However, it is worth noting that about half of the people with the most anti-Israel views show no evidence of antipathy toward the Jewish people.

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The last column of Table 6 shows summary statistics for respondents expressing the strongest agreement about Jews having too much influence. Their antipathy toward Jews is consistent across all the other measures about Jews, and their mean view about Israel being unjust is nearly half a standard deviation above the whole sample mean (64% express the strongest agreement which is double the percent for the whole sample). Relative to the whole sample, these respondents are much more right-wing, less likely to be college educated (nine percent relative to 29%), less religious, much older, have less favorable views about foreigners and Islam, and more skewed toward men (70% male).

5. OLS Analysis of the German General Social Survey Data

Table 7 presents an OLS analysis of a person's views about Israel as a function of how much the person agrees that "Jews have too much influence in the world"—the main treatment variable. The first column shows a strong, positive relationship between thinking that Jews have too much influence and believing that it is unjust for Israel to be taking Palestinian land, even after controlling for a person's age and gender. The following columns add controls to each successive specification such as: education, county-level percent of the population foreign-born, county unemployment rate, state of residence fixed-effects (17 states), fixed-effects for each level of the political spectrum (self-placement on the political spectrum from Left to Right, ranked 1 to 10 respectively), nine categories of religious affiliation (including non-affiliated), a dummy variable for self-reporting to be a "very proud German," views about whether "foreigners enrich the culture," and views about whether Islam fits in with German culture. The motivation for including these control variables comes from Kempf (2015), who argues that left-wing worldviews are correlated with antisemitism and anti-Israel sentiments. Beattie (2016) also shows that anti-Israel criticism and antisemitism are correlated with left-wing views about pacifism, human rights, a tendency to blame the more powerful side in a conflict, as well as right-wing ideology and Islamophobia. In addition, Zick et al. (2008) show that antisemitic views are part of a set of core prejudice beliefs against "out-groups" such as Muslims, homosexuals, women, and immigrants. Fortunately, the data set contains proxies for almost all of these factors: political ideology, religion, and views about nationalism, Muslims, and foreigners. Therefore, the first goal of the empirical analysis is to separate the effect of these other views from the effect of holding anti-Jewish sentiments.

The most noteworthy pattern in the upper panel of Table 7 is that the coefficient on "Jews have too much influence" is very stable when progressively adding more control variables. In fact, the estimated effect tends to get larger with more controls in the specification.[19] The bottom panel of Table 7 shows a similar pattern using the second

19. As discussed later in Table 9, thinking that Jews have too much influence is correlated with being on the right-wing of the political spectrum, being a "proud German," education, and negative views about foreigners and Islam.

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TABLE 7. OLS results for views on Israel and Jews have too much influence.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Israel—unjust to take Palestinian land (agreement scale 1–7)						
Jews—too much influence	0.118***	0.139***	0.144***	0.150***	0.149***	0.159***	0.162***
	(0.0180)	(0.0188)	(0.0189)	(0.0196)	(0.0197)	(0.0201)	(0.0204)
Observations	2,438	2,438	2,438	2,378	2,330	2,321	2,309
	Israeli policy justifies antisemitism (agreement scale 1–7)						
Jews—too much influence	0.389***	0.396***	0.398***	0.389***	0.386***	0.384***	0.376***
	(0.0204)	(0.0212)	(0.0213)	(0.0218)	(0.0220)	(0.0224)	(0.0226)
Observations	2,470	2,470	2,470	2,410	2,359	2,349	2,336
Age and age squared	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Gender	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education categories		Yes	Yes	Yes	Yes	Yes	Yes
County—percent foreigners			Yes	Yes	Yes	Yes	Yes
County—unemployment rate			Yes	Yes	Yes	Yes	Yes
State of residence			Yes	Yes	Yes	Yes	Yes
Political left–right spectrum categories				Yes	Yes	Yes	Yes
Religion categories				Yes	Yes	Yes	Yes
View—very proud german					Yes	Yes	Yes
View—foreigners enrich culture						Yes	Yes
View—Islam fits in with German culture							Yes

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each column and panel represents a separate regression. The sample is taken from the German General Social Survey (ALLBUS) from 2016 and is described in the notes to Table 6.

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TABLE 8. OLS results for various views about Jews.

	Israel—unjust to take Palestinian land (agreement scale 1–7) (1)	Israeli policy justifies antisemitism (agreement scale 1–7) (2)
Overall views on Jews—first factor	0.362***	1.186***
	(0.0478)	(0.0459)
Overall views on Jews—mean of five views	0.202***	0.762***
	(0.0310)	(0.0307)
Jews have too much influence	0.144***	0.398***
	(0.0189)	(0.0213)
Jews are not without blame for persecution	0.0897***	0.487***
	(0.0208)	(0.0220)
Jews take advantage of holocaust	0.146***	0.333***
	(0.0178)	(0.0199)
Jews—it would be unpleasant to have a Jewish neighbor	0.0649**	0.266***
	(0.0269)	(0.0295)
Jews—not ashamed of German crimes against Jews	−0.0367*	0.0875***
	(0.0202)	(0.0243)

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each coefficient represents a separate regression where the additional controls variables are the ones specified in column (3) of Table 7, and they include age, age-squared, gender, education categories, county percent foreign, county unemployment rate, and state fixed-effects. The sample is taken from the German General Social Survey (ALLBUS) from 2016 and is described in the notes to Table 6.

measure of a person's views about Israel as the outcome measure: How much they agree that "Israeli policy justifies antisemitism." The coefficient on views about Jews is unchanged when adding more and more personal and geographic controls. Overall, Table 7 shows that antisemitic tendencies are very significantly related to their views about Israel, even after controlling for many other ways of characterizing a person's view about the world, German society, German politics, foreigners, and Islam.

Table 8 examines whether these findings are robust to the other measures of anti-Jewish views. For both views about Israel, the previous results in Table 7 for the main measure of antisemitic beliefs (Jews have too much influence) are similar to the estimates in Table 8 obtained using the other views about Jews, including the two summary measures of all five views toward Jews (the mean and the first factor). The one exception is the estimated effect of "Not being ashamed of German Crimes against Jews" on one of the two views about Israel ("Unjust to take Palestinian Land").[20] But,

20. A principal component analysis of all five views shows this variable to be very different than the other four—the factor loading on the first factor is much smaller than the others, as noted above, and the

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the other measure of Israeli views for this treatment variable, as well as all the other ways of measuring antisemitic attitudes, show a robust effect pointing to anti-Israel sentiments increasing with antisemitic beliefs.

Online Appendix Table A.6 performs a similar analysis within subgroups of the population characterized by age, gender, and education. Within each subgroup, the same pattern emerges whereby the estimated effect is significantly positive, and increases in magnitude as additional controls are added to the specification. The estimates are remarkably similar for all age, gender, and education groups, and as will be discussed later during the IV analysis, the relationship between the treatment variable and the additional control variables differs between these groups. Therefore, the finding that the estimated effect gets larger as more controls are added for every group provides additional support for the idea that estimates do not suffer from an omitted variable bias away from zero.

Online Appendix Table A.7 uses the tools developed by Oster (2019) and Krauth (2016) to estimate the bounds on the treatment effect, if the treatment variable is no more correlated with unobservable factors than with the observed control variables. After allowing for this possibility, the table shows that the estimate is likely to be substantially larger than the OLS estimates in Tables 7 and 8, which is consistent with the larger effects found in the IV analysis below.

6. Instrumental Variables Analysis of the German General Social Survey Data

The previous section demonstrated that there is a strong relationship between holding anti-Jewish and anti-Israel views that is not explained by political leanings or beliefs about religion, foreigners, and Islam. These are indeed the main candidates that could generate a spurious correlation between views about Jews and Israel. However, there is always the potential for other omitted variables to create a bias, as well as reverse causality and measurement error. The latter issue could be significant if German respondents are uncomfortable with expressing their true views about Jewish people in a survey.

To be a valid instrument, we need a variable which affects anti-Jewish beliefs, and has no direct effect on views toward Israel. The motivation for the instrument comes from Voigtländer and Voth (2015), who argue that Nazi-indoctrination prior to and during World War II effectively brainwashed Germans to be antisemitic. They write: "Between 1933 and 1945, young Germans were exposed to antisemitic ideology in schools, in the (extracurricular) Hitler Youth, and through radio, print, and film. As a result, Germans who grew up under the Nazi regime are much more antisemitic than those born before or after that period." Evidence for this idea is displayed in Figure 1

uniqueness measure is much higher. These findings indicate that responses to this measure are not very consistent with the respondent's responses to the other four measures. One possible reason for this is that respondents who are not antisemitic at all may not be ashamed of German crimes against Jews because they were not alive when they occurred, and therefore, are not personally responsible for those crimes.

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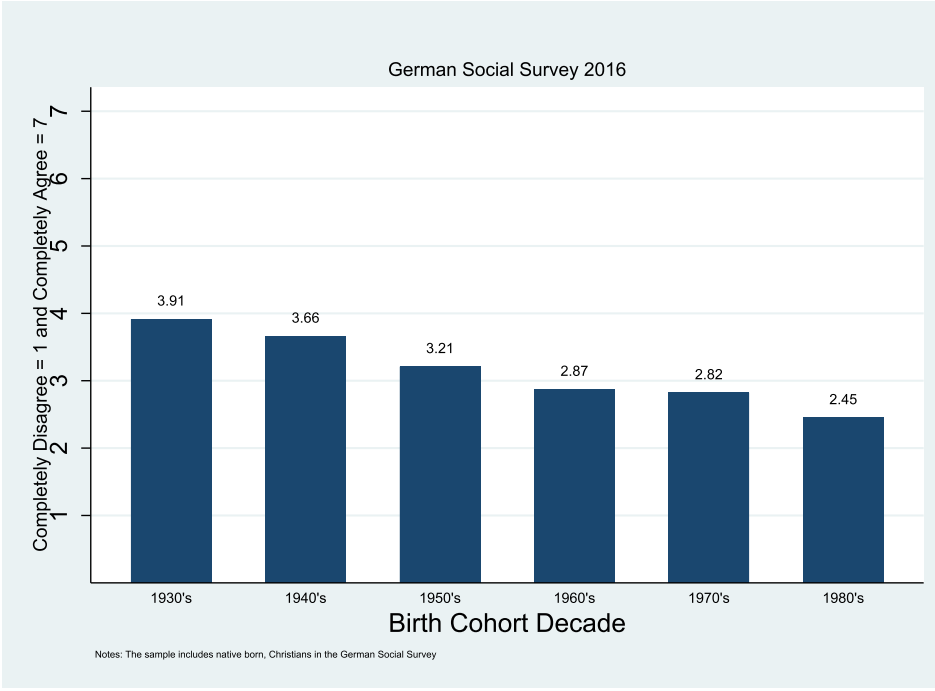


FIGURE 7. Jews have too much influence by birth cohort in 2016.

in Voigtländer and Voth (2015), which shows that individuals born in the 1930s and 1940s are much more antisemitic than those born after the 1940s.

Similar patterns are found in the sample used in our analysis in Figure 7. Individuals born in the 1930s and 1940s are much more likely to agree that Jews have too much influence than those born after, although the pattern is not as sharp as the one displayed in Voigtländer and Voth (2015). Table 9 shows that being born between 1930 and 1945 has a significant positive effect on believing that Jews have too much influence, even after controlling linearly for age. This finding is consistent with the argument in Voigtländer and Voth (2015) that children during this time were brainwashed to be antisemitic. Moreover, Israel did not exist during this period, so the indoctrination was specific to being anti-Jewish and not being anti-Israel.

With this motivation, the analysis uses being born between 1930 and 1945 as an instrument for attitudes toward Jews, in order to examine its effect on views about Israel. This strategy has the advantage that it is motivated by existing published work, and that year of birth is obviously exogenously determined. There are, however, a few potential issues that need to be addressed. First, as demonstrated by Table 9, the effect of being born between 1930 and 1945 ("the treatment period") has a significant, but not strong enough, effect on views about Jews once the specification controls linearly for age. In the third panel of Table 9, the results show that a much stronger effect of being born in the "treatment period" is found if we control for age in a more aggregate

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TABLE 9. Reduced form, first stage, and age patterns of personal views.

	Jews—Too Much Influence (1)	Political Spectrum (2)	Very Proud German (3)	No Religion (4)	College Graduate (5)	Foreigners Enrich Culture (6)	Islam Fits with German Culture (7)	Israel—Unjust to Take Palestinian Land (8)	Israeli Policy Justifies Antisemitism (9)
Age	0.0283***	−0.00255	0.00187***	−0.000201	−0.00526***	−0.0123***	−0.0168***	0.0154***	0.0348***
	(0.00240)	(0.00214)	(0.000576)	(0.000606)	(0.000548)	(0.00226)	(0.00211)	(0.00209)	(0.00241)
Cohort: born 1930–1945	0.227*	0.389***	0.0479	−0.0771**	0.0642**	−0.355***	−0.118	0.104	0.132
	(0.135)	(0.117)	(0.0310)	(0.0328)	(0.0278)	(0.126)	(0.115)	(0.116)	(0.133)
Age	0.0244***	−0.00922***	0.00105	0.00112	−0.00636***	−0.00628**	−0.0148***	0.0136***	0.0326***
	(0.00321)	(0.00283)	(0.000774)	(0.000829)	(0.000772)	(0.00307)	(0.00293)	(0.00285)	(0.00324)
Cohort: born 1930–1945	0.519***	0.241**	0.0439	−0.0668**	−0.0377*	−0.420***	−0.240**	0.337***	0.618***
	(0.118)	(0.101)	(0.0267)	(0.0280)	(0.0227)	(0.109)	(0.0954)	(0.0988)	(0.115)
Cohort: born 1930–1960	0.549***	−0.151**	0.0489**	0.0301	−0.105***	−0.156*	−0.420***	0.192**	0.583***
	(0.0869)	(0.0758)	(0.0199)	(0.0219)	(0.0191)	(0.0820)	(0.0770)	(0.0748)	(0.0880)
Jews—too much influence		0.156***	0.0248***	0.00762	−0.0584***	−0.252***	−0.246***	0.138***	0.431***
		(0.0192)	(0.00469)	(0.00509)	(0.00421)	(0.0191)	(0.0178)	(0.0175)	(0.0196)
Israel—unjust to take Palestinian land	0.191***	−0.0581***	0.00337	0.0123**	0.00409	0.0402*	−0.00847		0.454***
	(0.0240)	(0.0220)	(0.00563)	(0.00598)	(0.00533)	(0.0237)	(0.0221)		(0.0222)
Observations	2,585	2,694	2,718	2,795	2,795	2,777	2,746	2,531	2,561

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Every column and row panel (except for the row referring to the number of observations) presents coefficients from a separate regression with the column variable as the dependent variable and the row panel variable(s) as the independent variables. Each of these regressions also control for a gender dummy variable. The number of observations refers to the regressions in the first row panel. The sample is taken from the German General Social Survey (ALLBUS) from 2016 and is described in the notes to Table 6.

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way—controlling for being born before or after 1960. Specifically, this regression shows that once we control for someone being born before or after 1960, there is a strong effect on anti-Jewish sentiments of being born before 1945. That is, there is a strong difference between those born during the Nazi regime of 1930 and 1945, relative to individuals born between 1946 and 1960. In order to have an instrument strong enough by conventional standards in the first stage, this specification is used, but the sensitivity of the results to this assumption will be examined. In addition, a linear specification of birth year as an instrument for antisemitic views, which has a very strong first stage (the first column of Table 9), will be explored.

The second issue is the exclusion restriction assumption—that being born during the treatment period does not have a direct effect on views toward Israel. This issue is addressed in several ways. First, the analysis will control for being born before versus after 1960 in the first and second stage. That is, the IV analysis allows for a broad direct effect of age on views toward Israel, while assuming that there is no direct effect of being born between 1930 and 1945 relative to 1946 and 1960. The robustness of the results will also be examined by making this assumption even more restrictive—that there is no direct effect at all of age—and seeing whether the conclusions are sensitive to alternative, and even stricter, identifying assumptions.

The second way to address the validity of the exclusive restriction is by examining the sensitivity of the results to the inclusion or exclusion of a variety of other personal views about the world, which may be correlated with age. To be more precise, it is reasonable to assume that age itself does not have a direct effect on views toward Israel, but age is correlated with the types of views that people of a similar birth cohort tend to hold about the world, politics, religion, nationalism, Islam, foreigners, and Jews. It is likely that these other cohort-specific ideologies shape one's views about Israel as well. Table 9 supports this idea by showing that these other viewpoints, including those about Israel, are correlated with birth year (i.e., age). Individuals born in earlier cohorts tend to be more nationalistic, religious, less educated, and less amenable to foreigners, Islam, and Israel. Therefore, the IV analysis will examine what happens to the results when these "other views," which are correlated with age and likely have a direct effect on views toward Israel, are added to the IV specification. This sensitivity analysis will shed light on the extent to which the exclusion restriction assumption may be affecting the main conclusions.

Table 10 presents the second stage IV results from a regression of Israel views on our main measure of antisemitic beliefs (Jews have too much influence), which is instrumented with a dummy variable for being born between 1930 and 1945. The specification in the first column has no other control variables, except for gender. The coefficients for both outcome measures of opinions toward Israel are highly significant, and suggest a strong effect of being antisemitic on being anti-Israel. As noted above, the specification in column (1) assumes no direct effect of being born during the treatment period (1930–1945), relative to later cohorts, on views toward Israel. To relax this assumption to a large extent, the specification in the second column of Table 10 includes an indicator for being born before or after 1960. The results in column (2) are in fact larger and still statistically significant after relaxing,

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TABLE 10. IV results for views on Israel and Jews have too much influence.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Israel—Unjust to take Palestinian Land (Agreement Scale 1–7)							
Jews—too much influence	0.489***	0.589***	0.830**	0.850**	0.914**	0.854**	0.934**	0.981**
	(0.111)	(0.227)	(0.330)	(0.339)	(0.358)	(0.363)	(0.382)	(0.401)
Cohort dummy—born 1930–1960		−0.134	−0.114	−0.115	−0.193	−0.173	−0.184	−0.161
		(0.185)	(0.199)	(0.198)	(0.221)	(0.229)	(0.229)	(0.220)
	Israeli Policy Justifies Antisemitism (Agreement Scale 1–7)							
Jews—too much influence	1.120***	1.204***	1.434***	1.409***	1.410***	1.405***	1.466***	1.528***
	(0.136)	(0.286)	(0.414)	(0.413)	(0.431)	(0.450)	(0.471)	(0.513)
Cohort dummy—born 1930–1960		−0.114	−0.0813	−0.0674	−0.123	−0.138	−0.131	−0.139
		(0.240)	(0.258)	(0.252)	(0.280)	(0.298)	(0.294)	(0.294)
Gender	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education categories			Yes	Yes	Yes	Yes	Yes	Yes
County—percent foreigners				Yes	Yes	Yes	Yes	Yes
County—unemployment rate				Yes	Yes	Yes	Yes	Yes
State of residence				Yes	Yes	Yes	Yes	Yes
Political left–right spectrum categories					Yes	Yes	Yes	Yes
Religion categories					Yes	Yes	Yes	Yes
View—very proud German						Yes	Yes	Yes
View—foreigners enrich culture							Yes	Yes
View—Islam fits in with German culture								Yes

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each row panel and column represents a separate IV regression. The instrument for the endogenous variable (Jews—too much influence) is an indicator for being born between 1930 and 1945. The sample is taken from the German General Social Survey (ALLBUS) from 2016 and is described in the notes to Table 6.

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to a meaningful extent, the assumption that age has no direct effect on attitudes toward Israel. This finding suggests that the more restrictive identifying assumption in column (1) is leading to a bias toward 0, if it creates a bias at all. Moreover, the coefficient for the direct effect of the 1930–1960 birth-cohort in column (2) is insignificant, which is consistent with the identifying assumption in column (2) that age within broad cohorts does not have a direct effect.[21]

The remaining columns in Table 10 add more and more controls to the specification. Again, many of these controls account for the typical ways in which older Germans have a different worldview about their country, politics, religion, foreigners, and Islam. In addition, Table 10 adds other variables that may have a direct effect and could be correlated with age such as geographic controls, the local unemployment rate, and levels of education. In the final column of Table 10, the identifying assumption is that—conditional on the person's views about foreigners, German pride, Muslims, religion, as well as geographic controls and broad cohort effects—being born between 1930 and 1945 relative to being born between 1946 and 1960 has no direct effect on views regarding Israel.

Going across the columns in Table 10 reveals a very striking pattern whereby adding more control variables yields larger estimates that remain statistically significant. That is, adding controls for a direct effect of all the measures regarding a person's worldview that are correlated with age only makes the estimated effect of antisemitic beliefs on views about Israel stronger. It is important to note that the specifications, which add more and more personal views is in addition to allowing for a direct overall effect of the age cohort (pre- and post-1960). And, the estimated direct effect of being born before 1960 is always insignificant, which is consistent with the identifying assumption that being born between 1930 and 1945 relative to being born between 1946 and 1960 has no direct effect on views regarding Israel.

The robustness of the results to the inclusion or exclusion of these direct controls for age and worldviews that are correlated with age, suggests that the omission of unobserved views, which may be correlated with age is not biasing the results away from zero. This kind of sensitivity analysis is used in the IV literature to support a causal interpretation (see Alesina, Giuliano, and Nunn 2013 and Autor, Dorn, and Hanson 2013).[22] In fact, the results in Table 10 suggest that the addition of further omitted factors into the specification would only make the estimates stronger.[23]

21. The idea that the impact of Nazi indoctrination should be stronger for those born closer to 1930 relative to those born in 1945 and later is essentially the basis for using "birth year" as an IV after restricting the sample to those born between 1930 and 1959. This analysis is presented in Table 12. However, it is worth noting that our main IV results are not affected by deleting respondents born between 1940 and 1945 from the sample. For example, instead of the reported coefficient of interest in column (4) in the top panel of Table 10 of 0.85 (0.34), the estimate in the sample without those birth cohorts is 0.90 (0.39).

22. For example, see page 520 in Alesina, Giuliano, and Nunn (2013) and the discussion of Table 3 in Autor, Dorn, and Hanson (2013).

23. This is supported by the finding that the most restrictive exclusion restriction assumption in this setup, which assumes no other direct effect of anything correlated with age on views toward Israel (column one), yields the smallest estimates. Relaxing this assumption further and further across specifications in Table 10

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Regarding the estimated effects of the other control variables (not reported in Table 10), criticism of Israel increases with education, and with holding a positive view about how foreigners contribute to German culture. The magnitudes of these estimates are small relative to the estimated effect of views about Jewish people in Table 10. For example, relative to those without a college degree, obtaining a college degree increases criticism of Israel by 0.68, which is less than each step in the 1–7 scale of views regarding whether Jews have too much influence (the estimates in Table 10 are around 0.85 for each step). Each step in the 1–7 scale of holding a more favorable view of foreigners in Germany increases criticism of Israel by 0.15, compared with the estimate of 0.85 for each step regarding Jews. Somewhat surprisingly, insignificant effects are found for gender, religiosity, being a proud German, placement on the political spectrum, and sentiments toward Islam.

Table 11 examines the robustness of the IV results to the different ways of measuring the treatment variable—attitudes toward Jews. Table 11 also uses the two summary variables using all five measures of the respondent's beliefs about Jews. Similar to the OLS analysis, the results are robust across almost all measures of antisemitic beliefs (except the last one), including the two summary measures of antisemitism (first two rows). This is also true for both ways of measuring the outcome variable: the two different questions in the ALLBUS survey regarding opinions toward Israel.

The sample is too small to conduct a similar IV analysis within various subgroups (gender and education), given the limited number of respondents born between 1930 and 1945 in the survey. However, the evidence that age has no direct effect on attitudes toward Israel, as indicated by the insignificant coefficients on being born between 1930 and 1960 in Table 10, suggests that age (i.e., birth year) is likely to be a valid instrument for antisemitic beliefs. While it is true that being born during the Nazi era is associated with a spike in antisemitic views even many decades later (Voigtländer and Voth 2015), it is also the case that older cohorts are generally more antisemitic—even for those born after 1960 (see Online Appendix Table A.8). If we assume that age has no direct impact on Israel views for those born before 1960 and within those born after 1960 (but allow for the differences between those groups to have a direct effect), an IV analysis can be performed within various subgroups—with the advantage of having a very strong first stage.

Table 12 presents the results within broad age groups using birth year (age) as an instrument for antisemitic beliefs. For those born before 1960 and for those born after 1960, the findings are remarkably similar to the previous findings in Table 10 using the whole sample and a different IV strategy. The estimates are stronger for the younger cohort that is consistent with the idea that younger cohorts are more cautious and politically correct about answering sensitive questions about Jews and minority groups—so that a unit change in the response to these questions represents a bigger change in anti-Israeli sentiment for younger people. Once again, the estimates for

yields stronger and stronger estimates. This does not prove that the untestable exclusion restriction is satisfied, but provides support that it is not biasing the analysis toward finding a spuriously large, significant effect.

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TABLE 11. IV results for various views about Jews.

	Israel—unjust to take Palestinian land (agreement scale 1–7) (1)	Israeli policy justifies antisemitism (agreement scale 1–7) (2)
Overall views on Jews—first factor	2.178**	3.657***
	(0.889)	(1.125)
Overall views on Jews—mean of five views	2.262**	2.982***
	(1.076)	(0.972)
Jews have too much influence	0.850**	1.409***
	(0.339)	(0.413)
Jews are not without blame for persecution	0.799**	1.470***
	(0.351)	(0.426)
Jews take advantage of holocaust	1.635*	2.268**
	(0.947)	(1.029)
Jews—it would be unpleasant to have a Jewish neighbor	6.321	6.934
	(8.626)	(7.038)
Jews—not ashamed of German crimes against Jews	−1.742**	−2.553**
	(0.886)	(1.282)

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each coefficient represents a separate IV regression where the additional controls variables are the ones specified in column (4) of Table 10, and they include age, age-squared, gender, education categories, county percent foreign, county unemployment rate, and state fixed-effects. The instrument for the endogenous variable indicated in each row is an indicator for being born between 1930 and 1945. The sample is taken from the German General Social Survey (ALLBUS) from 2016 and is described in the notes to Table 6.

both groups get stronger as additional controls are added to the specification. This pattern is notable because the treatment variable (antisemitic views) is correlated with the additional control variables in a differential way across the two age groups. In particular, for the younger cohort, antisemitic views have a stronger relationship with being right-wing politically, proud to be German, religious, less educated, and anti-Islam (see Online Appendix Table A.8). Given that the treatment variable is correlated in markedly different ways with these other variables across groups, it is notable that the addition of these controls has a similar, strengthening effect on the estimates for both groups. Again, this robust pattern supports the identifying assumption that the instrument, year of birth, is not correlated with omitted variables in a manner that creates an upward bias.

Moreover, the specification in the upper panel of Table 12 for the older cohort (individuals born prior to 1960) can be considered an alternative way of modeling the instrument, used previously in Tables 10 and 11, to capture the long-term impact of Nazi indoctrination described in Voigtländer and Voth (2015). Consistent with the idea that a longer exposure to Nazi brainwashing yields a bigger impression on a

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TABLE 12. IV results by birth year cohorts (with birth year as the instrument).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Israel—unjust to take Palestinian land (agreement scale 1–7)						
Cohorts: birth years 1930–1959							
Jews—too much influence	0.615***	0.892***	0.884***	0.975***	0.952***	1.034***	1.088***
	(0.209)	(0.329)	(0.317)	(0.340)	(0.358)	(0.379)	(0.411)
Cohorts: birth years 1960–1990							
Jews—too much influence	1.004***	2.425	2.005*	1.470**	1.391**	1.210**	1.345**
	(0.372)	(1.534)	(1.105)	(0.668)	(0.604)	(0.490)	(0.539)
Cohorts: birth years 1930–1990							
Jews—too much influence	0.799***	1.432***	1.383***	1.258***	1.246***	1.222***	1.311***
	(0.193)	(0.437)	(0.414)	(0.353)	(0.356)	(0.332)	(0.357)
Cohort dummy—born 1930–1959	−0.296*	−0.451*	−0.405	−0.394*	−0.409*	−0.347	−0.330
	(0.170)	(0.272)	(0.253)	(0.234)	(0.240)	(0.217)	(0.215)
Gender	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education categories		Yes	Yes	Yes	Yes	Yes	Yes
County—percent foreigners			Yes	Yes	Yes	Yes	Yes
County—unemployment rate			Yes	Yes	Yes	Yes	Yes
State of residence			Yes	Yes	Yes	Yes	Yes
Political left–right spectrum categories				Yes	Yes	Yes	Yes
Religion categories				Yes	Yes	Yes	Yes
View—very proud German					Yes	Yes	Yes
View—foreigners enrich culture						Yes	Yes
View—Islam fits in with German culture							Yes

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each row panel and column represents a separate IV regression. The instrument for the endogenous variable (Jews—too much influence) is the respondent's year of birth. The sample is taken from the German General Social Survey (ALLBUS) from 2016 and is described in the notes to Table 6.

person's lifetime outlook, Online Appendix Table A.8 shows that the relationship between birth cohort and antisemitic beliefs is much steeper for the older cohort (born prior to 1960) relative to the younger one (post-1960). The age gradient in Online Appendix Table A.8 is nearly twice as big for the older cohort (0.036 vs. 0.019). Therefore, the linear specification of birth year as the instrument in the upper panel of Table 12 for the older cohort can be considered a sensible alternative strategy to the one used in Tables 10 and 11, which specified Nazi influence as a discrete jump in antisemitic beliefs for birth years prior to 1945 versus post-1945. This alternative specification of the Nazi indoctrination effect has a very strong first stage (the first column of Online Appendix Table A.8) and yields a very similar pattern of results to those obtained in Tables 10 and 11. Therefore, the overall findings are not sensitive to how the instrument is defined to capture the long-run impact of Nazi brainwashing for those most exposed to it.

In the bottom panel of Table 12, the IV analysis is performed with the whole sample but uses "year of birth" as the instrument, while controlling for the direct

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effect of being born before or after 1960. The results are somewhat stronger than those in Table 10, which used the instrument based on the discrete comparison between those born in 1930–1945 and 1946–1960. Again, the estimates get stronger as more controls are added, and the direct effect of being born before 1960 is either insignificant or negative—indicating a more positive view of Israel despite the older group having more negative views about Jews and Israel according to the summary statistics in Online Appendix Table A.8.

In Online Appendix Table A.9, similar findings are found within subgroups of men, women, college graduates, and non-college graduates. The only exception is that the estimates are not consistently statistically significant for college graduates—the estimates are not precise, which may be due to a much smaller sample size relative to the other groups. As shown in Online Appendix Tables A.10 and A.11, there are differences in the way the treatment variable is correlated with the additional control variables within each subgroup, so it is again notable that the second-stage estimates are consistently stronger as more controls are added for each subgroup.

Overall, the IV results confirm the general findings of the OLS analysis that antisemitic beliefs lead to an anti-Israel perspective. It is worth noting that the estimated IV effects are considerably larger than the OLS coefficients. The IV estimate in the first column of Table 11 for the summary measure of all five views using factor analysis is 2.178 relative to a coefficient of 0.362 using OLS in Table 8. The predicted effect on Israel views of a one standard deviation change in this treatment variable (0.82 in Table 6) is 1.786 for IV and 0.297 using OLS. Compared to a one standard deviation change in this outcome variable (1.62) in Table 6, the IV coefficient predicts a change in the outcome variable of 1.10 standard deviations. The predicted effect using the OLS coefficient in Table 8 is 0.18 of a standard deviation. While both coefficients are not small in magnitude, the IV coefficient is considerably larger. Given the sensitive nature about asking people, particularly German citizens, about their views toward the Jewish people, the most likely reason for OLS being biased downwards would be that some people are reluctant to reveal anti-Jewish beliefs in a survey, and thus, measurement error in this variable could be prevalent. This idea is consistent with the experimental evidence in Cohen et al. (2009), where subjects revealed stronger antisemitic sentiments under more pressure to tell the truth. In addition, measurement error may be the result of differences across respondents in the way they interpret the statements about Israel and Jews in the survey, and in how they translate their views into subjective rankings.

Empirical evidence for the idea that respondents may be reluctant to express negative views about Jews in surveys is supported by looking at how these responses vary with the characteristics of the interviewer. In our main sample, responses to the question about whether Jews have too much influence are 0.22 points higher for respondents who had an interviewer at least 75 years old (in the 2016 survey). These are interviewers that were born in 1941 or earlier, which is consistent with the idea that survey respondents may feel more comfortable revealing negative opinions about Jews with interviewers who are less likely to be judgmental about such views. (The age of the interviewer is not correlated with the age of the respondent.)

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Online Appendix Table A.12 presents results using the age of the interviewer as an IV for the interviewee's views about Jews, and reveals strikingly similar results, in terms of size and significance, to the main IV results using the birth cohort of the interviewee in previous tables. However, it should be noted that the first-stage F-statistic is about 5, which is not strong enough by conventional standards (only 10% of the sample was interviewed by someone 75 years old or older).

Overall, the IV results suggest that views about Israel are very influenced by attitudes toward Jews. It is natural to ask how much anti-Israel sentiment is determined, or not determined, by personal biases against Jews. Given the imperfect nature of the questions in the survey, along with the possibility that people may hide their true feelings regarding sensitive questions, it would take very strong assumptions to arrive at a reliable estimate of this. However, to give a sense of this picture, we use the estimates to compute the counterfactual distribution of views about Israel if everyone showed at least some disagreement with the idea that Jews have too much influence in the world (i.e., capping the response at 3 in Figure 4).

This counterfactual exercise lowers the mean response to the question about Israel being "unjust" by 0.76, compared with the mean and standard deviation of 5.30 and 1.62, respectively. The change in the mean is about half of a standard deviation (47% of 1.62), which represents a significant shift toward more favorable views of Israel. Capping the response to this question at "completely disagreeing" with the idea that Jews have too much influence (a value of "1," which is the mode response to this question as seen in Figure 4) reduces the mean view about Israel to 3.50, which represents a shift in the mean of 1.8 compared with a standard deviation of 1.62. Therefore, forcing everyone to have the most favorable view of Jews (which is also the most frequent view) shifts the distribution of views about Israel dramatically in a favorable direction, but this counterfactual distribution still contains 18.5% of the sample with a fairly negative view about Israel being "unjust" of at least 6, compared to 52% in the actual survey responses. Who are these people that still have such a negative view of Israel in this exercise? Unsurprisingly, they are the type of people those display little negativity toward Jews alongside hostility to Israel (as discussed regarding Table 6 above). Relative to those with a counterfactual view of Israel below a value of 6 after capping the responses about Jewish influence to a value of 1: They are much more likely to be college educated (42% vs. 25%), more left-wing, less likely to be a "very proud German," and with more favorable views of foreigners and Islam. It is worth noting that only 10.2% of the sample would hold the most negative view of Israel (a value of 7) in this experiment, compared with 32.7% in the actual survey responses. Therefore, the analysis does not reveal any evidence that one-third of the sample respondents with the most negative view of Israel are motivated by anti-Jewish sentiment.

7. Robustness to Using Data from Other European Countries

In this section, the robustness of the results from the German General Social Survey is examined with the data used by Kaplan and Small (2006) from 10 European countries.

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The ADL commissioned a survey in 2004 to gather information about attitudes of Europeans regarding Jews and other minority groups, as well as their degree of social contacts with these groups. The survey also asked questions about views toward Israel.

Online Appendix Table A.13 presents the summary statistics by country for the main variables of interest regarding attitudes toward Jews, Israel, and other minority groups (immigrants). The main variable for anti-Jewish sentiment used in this analysis is the "antisemitic" index created by Kaplan and Small (2006). This index is a summary measure of 11 different questions that respondents were asked regarding Jews, such as whether Jews have too much power in the business world, in financial markets, and in their own country. The other questions are typically about whether Jews are honest, irritating, too loyal to Israel, uncaring about other people, stick together too much, or are too shrewd in business. Kaplan and Small (2006) created an "antisemitic index", which is based on whether the respondent expressed a negative sentiment about Jews on at least 5 of the 11 questions. They showed that their results are not particularly sensitive to this definition.

The "anti-Israel index" constructed by Kaplan and Small (2006) is based on four questions in the ADL survey about the respondent's views toward Israel. In particular, the survey asked respondents about whether they regard Israel as an apartheid state, whether they view Israel as being more responsible for violence during the Second Intifada relative to the Palestinians, whether they think Israel targets civilians in military operations, and whether there is any justification for Palestinian suicide bombers. The "anti-Israel index" created by Kaplan and Small (2006) is the number of these questions that the respondent agreed to.

In addition to using the variables constructed by Kaplan and Small (2006) from the ADL survey, a summary measure using factor analysis on all the views toward Jews was created, as well as a similarly constructed measure for antisemitic sentiments. Online Appendix Table A.13 displays the means of these variables across 10 European countries. The survey contains respondents between the ages of 18 and 64 as of 2004, so the earliest year of birth is 1940. The sample is restricted to Christian and religiously non-affiliated respondents, for reasons stated above regarding the German General Social Survey. Figure 8 reveals a strong positive relationship between antisemitism and holding anti-Israel views in the ADL survey.

Table 13 presents the OLS results from regressing the anti-Israel index on the antisemitic index after controlling for age, gender, and country fixed-effects. The positive coefficient is consistent with the results from the German General Social Survey, suggesting that antisemitic views have a positive impact on anti-Israel views. Controls for religion, income groups, views about "commonality with other races and religious groups," and views about immigrants are added sequentially across columns. Similar to the previous results with German data, the OLS coefficient increases in magnitude as more control variables are added to the specification.

In the analysis of the German Social Survey, the instrumental variable was motivated by the idea that the Nazi indoctrination of Germans born during World War II created lasting anti-Jewish attitudes. Since the ADL survey includes many European countries, and does not contain respondents born before the 1940s, this

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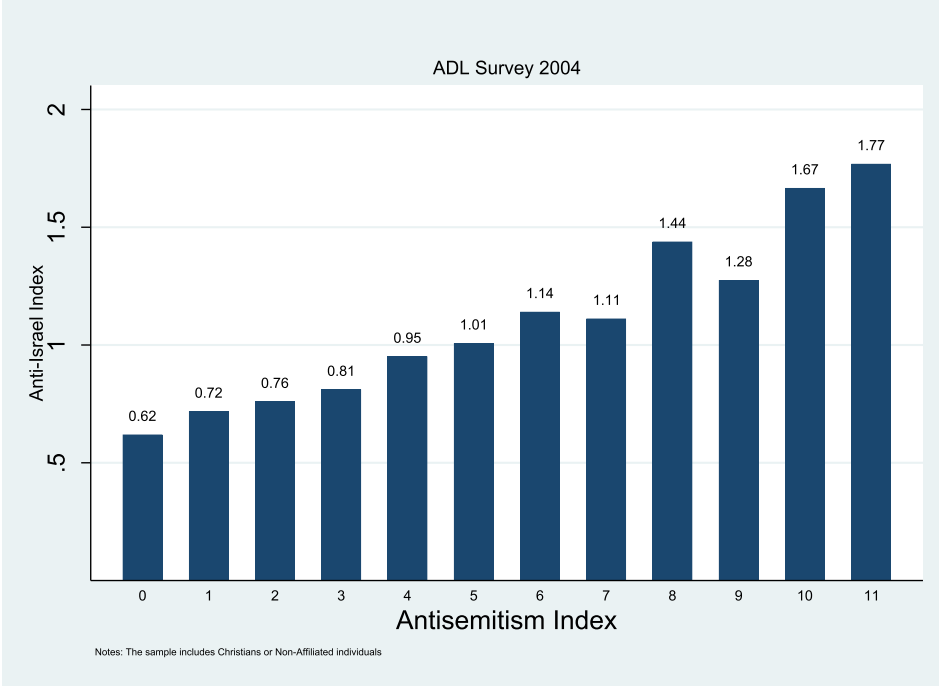


FIGURE 8. Views on Israel by views on Jews.

strategy does not directly extend to the sample used in Table 13. However, in the previous section, the evidence suggested that "year of birth" (age) is a valid instrument for antisemitic attitudes. Older cohorts in the year 2004 are typically more nativist and antisemitic, and while this pattern may be more pronounced in Germany due to the Nazi brainwashing of children during the 1930s and 1940s, it is also a more general pattern across European countries (and in Germany for those born after the Nazi era as seen in Online Appendix Table A.8).

Evidence for this idea is presented in Figure 9 and in Online Appendix Table A.14. The first column of the table shows that the antisemitic index is significantly higher for individuals born in the 1940s and 1950s compared with later cohorts, even after controlling for gender and country fixed-effects. The next two columns demonstrate two patterns: (1) the main dividing point is that those born before 1960 are more antisemitic than those born after; and (2) within those born before 1960, those who were born before 1950 are much more antisemitic than those born during the 1950s. These findings are consistent with the idea behind the two IV strategies in the previous analysis of the German Social Survey—that older individuals are more antisemitic, and this is true even within those born before 1960. (The ADL data do not contain the precise age of respondents, only the decade of birth.)

Online Appendix Table A.14 shows similar cohort effects for other personal views regarding whether the person feels commonality with other racial and religious

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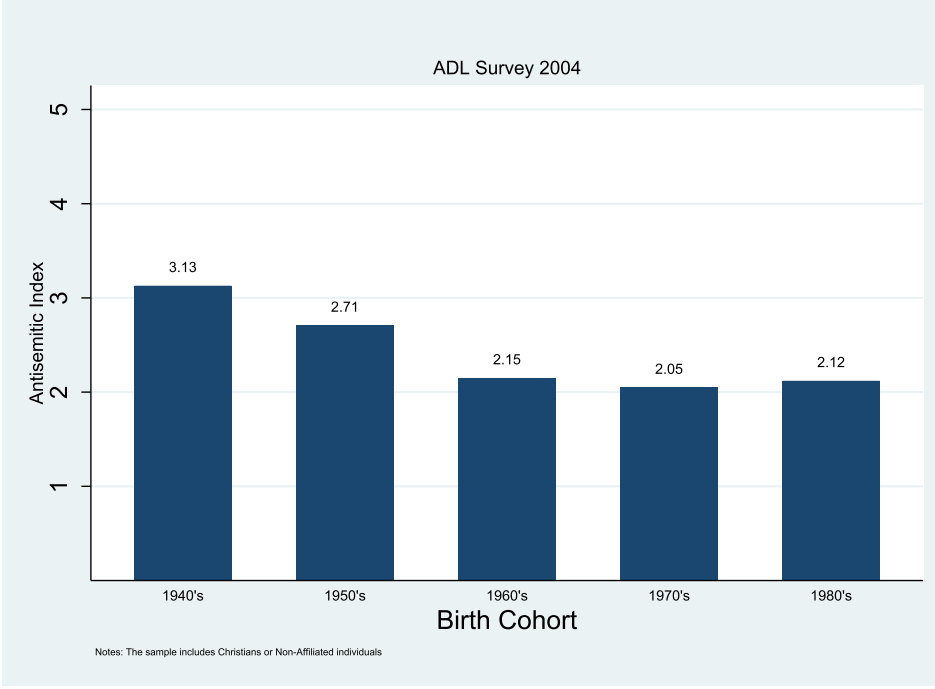


FIGURE 9. Views on Jews by birth cohort.

groups, whether immigrants are a burden on society, identifying as Christian, and anti-Israeli sentiments. Some of these coefficients are not statistically significant, but they point to a general pattern whereby individuals born during the 1940s are more close-minded about Jews, immigrants, other religious and racial groups, and Israel. That is, individuals born during 1940s are more insular and grew up in a world where antisemitic beliefs were more common and open relative to later cohorts.

Consistent with the previous evidence that birth year has no direct effect on views toward Israel, the first column of the IV results in Table 14 includes no other controls except for gender and country fixed-effects. The estimate is positive, significant, and larger in magnitude than the OLS estimates. (The IV coefficient is 0.134 compared with the largest OLS estimate of 0.093 in Table 13.) This specification assumes that age has no direct effect at all on the outcome variable, whereas the second column relaxes this assumption by including a control for being born before 1960. The coefficient estimate increases in the second column to 0.233 from 0.134.

In the remaining columns of Table 14, more control variables are adding sequentially for income, sentiments regarding commonality with other racial and religious groups, and views toward immigrants. Again, these variables reflect how the worldview of people born during the 1940s differs from those born in the 1950s and later cohorts. Similar to our previous findings, adding more control variables produces larger and statistically significant effects of antisemitic beliefs on anti-Israel views.

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TABLE 13. ADL survey—OLS analysis of the anti-Israel index.

	Anti-Israel index					
	(1)	(2)	(3)	(4)	(5)	(6)
Antisemitic index	0.0855***	0.0847***	0.0856***	0.0856***	0.0874***	0.0931***
	(0.00744)	(0.00745)	(0.00744)	(0.00738)	(0.00746)	(0.00746)
Age (decade)	−0.000548					
	(0.00121)					
Birth cohort—1940s		−0.0166	0.00382	−0.0332	−0.0302	−0.0423
		(0.0571)	(0.0573)	(0.0577)	(0.0575)	(0.0574)
Birth cohort—1950s		−0.0746	−0.0606	−0.107**	−0.112**	−0.120**
		(0.0533)	(0.0534)	(0.0537)	(0.0535)	(0.0532)
Birth cohort—1960s		−0.0882*	−0.0807	−0.126**	−0.132**	−0.147***
		(0.0518)	(0.0520)	(0.0527)	(0.0524)	(0.0523)
Birth cohort—1970s		−0.0247	−0.0241	−0.0729	−0.0857	−0.104*
		(0.0542)	(0.0542)	(0.0549)	(0.0547)	(0.0544)
Gender	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Christian			Yes	Yes	Yes	Yes
Income group effects				Yes	Yes	Yes
Commonality with others					Yes	Yes
Views on immigrants						Yes
Observations	3,859	3,859	3,859	3,859	3,859	3,859

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each column represents a separate regression. The sample is taken from the ADL Survey and is described in the text and in the notes for Online Appendix Table 13.

In the bottom panel of Table 14, a similar analysis is conducted with the summary measures for antisemitism and anti-Israel beliefs using factor analysis on all the individual questions regarding these two issues in the ADL survey. These summary variables are used instead of the indices created by Kaplan and Small (2006). Again, a very similar pattern emerges whereby adding more controls leads to larger estimates.

It is worth noting that, similar to the analysis with the German Social Survey, the most restrictive identifying assumption in the IV analysis yields the smallest estimates, and relaxing this assumption increases the estimated coefficient substantially in magnitude. In other words, regarding concerns that being born in the 1940s relative to the 1950s is correlated with factors which directly affect views about Israel, the analysis shows that controlling for several of these factors strengthens the estimated effects. In addition, the direct effect for age in the analysis, the dummy variable for being born in the 1940s or 1950s, is not significant in any specification. These results are consistent with the idea that being born in the 1940s relative to being born in the 1950s has no direct effect on Israel views, and that controlling even further for other factors correlated with age would only yield larger estimates.

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TABLE 14. ADL survey IV results.

	(1)	(2)	(3)	(4)	(5)	(6)
	Anti-Israel Index					
Antisemitic index	0.134***	0.233*	0.253*	0.282**	0.354*	0.354*
	(0.0474)	(0.126)	(0.130)	(0.138)	(0.182)	(0.182)
Birth cohorts–1940s and 1950s		−0.115	−0.115	−0.142	−0.161	−0.161
		(0.103)	(0.105)	(0.112)	(0.130)	(0.130)
	Overall Anti-Israel (First Factor)					
Overall antisemitic (first factor)	0.301***	0.531**	0.571**	0.635**	0.817**	0.817**
	(0.0897)	(0.262)	(0.271)	(0.292)	(0.411)	(0.411)
Birth cohorts–1940s and 1950s		−0.0927	−0.0936	−0.115	−0.136	−0.136
		(0.0751)	(0.0769)	(0.0838)	(0.104)	(0.104)
Gender	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Christian			Yes	Yes	Yes	Yes
Income group effects				Yes	Yes	Yes
Commonality with others					Yes	Yes
Views on immigrants						Yes

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Source: Each row panel and column represents a separate IV regression. The instrument for the endogenous variable ("Antisemitic Index" in the upper panel, and "Overall Antisemitic (First Factor)" in the lower panel) is an indicator for being born in the 1940s. The sample is taken from the ADL Survey and is described in the notes to Online Appendix Table 13.

In terms of the magnitude of the coefficients, the coefficient in the middle of the range in column (4) of Table 14 predicts that a one standard deviation change in the antisemitic index (2.36) would increase the anti-Israel index by 0.67 which is almost three-quarters of a standard deviation change (the standard deviation of the anti-Israel index is 0.922). Overall, the general pattern of the results across the OLS and IV analyses, including the magnitudes of the coefficients, are strikingly similar to the analysis with the German Social Survey.

8. Conclusion

Using two distinct empirical strategies with data from social media, the German General Social Survey, and a European survey conducted for the ADL, the evidence in this study suggests that critical stances about Israel are significantly influenced by negative views about Jewish people. These findings cannot determine whether a specific person who is critical of Israel is indeed antisemitic. However, the consistent pattern across each analysis showing much larger effects for IV relative to OLS suggests that some people are reluctant to express their true negative feelings about Jewish people, but feel more comfortable revealing these motivations in the form of anti-Israel expressions.

The importance of personal biases, and how they shape political views, has been examined in the literature regarding immigration policy, welfare policy, and the War

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on Terror. Other areas where these issues may be relevant include policies regarding legal protections against discrimination (gender, racial, religious, etc.), affirmative action, police funding and practices, drug legalization, voting laws, housing policies, or any other policy which one group perceives will have a disproportionate impact on another group. Differences in policy preferences regarding these issues will exist even without personal biases and prejudices. But, the extent to which discussions around these issues are colored by motives unrelated to the policies themselves could have a negative impact on conducting an honest public debate, and present barriers to enacting effective strategies to tackle important issues.

This may be particularly relevant when individuals do not even realize that they are biased. According to the psychology literature, individuals are sometimes affected by a "confirmation bias" (Kayman 1995), which leads them to pay more attention to information that confirms their original beliefs. Such an effect is likely to be compounded by the "echo chamber" encouraged by modern algorithms of news consumption and social media (Cinelli et. al. 2021). According to cognitive dissonance theory (Festinger 1957), individuals may also adopt positions on contentious issues in order to align their views with that of their identity (Gould and Klor 2019), rather than objectively assess both sides of an argument. In these types of situations, a rational discussion of facts and arguments may not be possible, which may lead to a discounting of views that are suspected to be influenced by biases.

In the epistemology literature, there are competing strands of thought on that issue. "Reliabilism" claims that we can only know something if the chain of sources is reliable (Goldman 2012). Therefore, if the psychological process underlying a person's belief is not reliable, the belief itself is considered unreliable. However, the principle of "Epistemic Injustice" (Fricker 2007) claims that testimony should not be discounted on the basis of group membership, and that the validity of a person's views should be evaluated independently of whether bias is suspected. In the theory of Bayesian persuasion (Kamenica 2019), a biased individual may strategically moderate their views in order to appear more reliable or even unbiased (Dziuda 2011). Therefore, even if an individual is suspected to be biased, the individual's views should not necessarily be ignored.

Insights from these fields highlight how complicated it is to assess the views of biased individuals, and conduct rational public debates on sensitive issues. This is particularly problematic when some people may hold a view because of a certain prejudice, while others with similar views are perhaps falsely accused of biases. The findings in this paper suggest that this is indeed the case regarding the Israeli–Palestinian conflict, and that one of the oldest forms of discrimination in the world, antisemitism, is still a significant factor in one of the most highly debated and intractable issues in international affairs.

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Supplementary Data

Supplementary data are available at *JEEA* online.

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